

KITEWRIGHT LIFT · BUILD PLAN

Kitewright Lift prototype build plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	KWL-BLD-001	Version	0.3
Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-lift		
Source	docs/05-build-plan.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-04	Amish Chadha	Amish's round-3 decisions 8A, 9A and 10A (KWL-DDR-004); ColdCell's packs as drawn on a re-sized, raised 2 mm deck; the Kitewright Core hung under the hub to the family envelope; payload shoes; AS150 everywhere; warm-weather operating note; sections 1, 2, 3.1, 3.3, 3.4, 3.5, 3.14 to 3.19, 3.22, steps 2, 3, 6, 11 to 15, first checks and safety stops
0.2	2026-10-03	Amish Chadha	Amish's decisions 34A and 35A (KWL-DDR-003); lighter hinge blocks, root fittings and motor clamps, lithium-ion ColdCell packs, frame-to-Core power leads (Core decision 33B)
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (KWL-DDR-002)

Kitewright Lift prototype build plan

Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register (docs/06-design-decisions.md), not here.

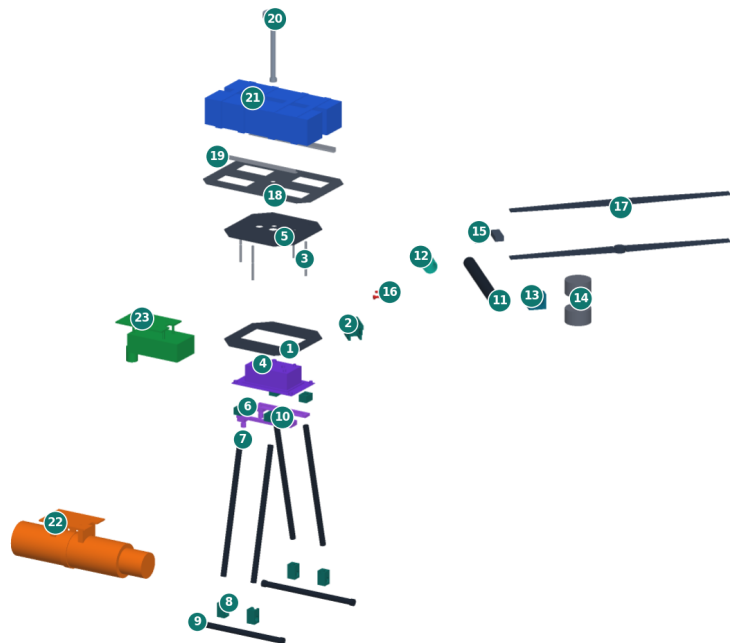
Safety: This plan builds a 25 kg class aircraft with eight 30 inch propellers, lithium-ion packs and a 400 V tether. Propellers can kill, lithium-ion packs can burn fiercely, and the tether carries a dangerous voltage. Every safety stop in section 6 is a hard stop: work does not go on until its conditions are true. Fly only where local rules allow, never near uninvolved people, and never carry or lift a person.

1. What you are building

Kitewright Lift prototype: every component in build order

Hub stack and gear first, the Core under the hub, then the arms (one of four shown), deck, packs and payloads

- 1 Bottom hub plate
- 2 Arm hinge blocks (4), one shown
- 3 Hub spacers and deck standoffs (8)
- 4 Kitewright Core (Core), hung under the hub
- 5 Top hub plate
- 6 Gear top blocks (4)
- 7 Gear struts (4)
- 8 Skid blocks (4)
- 9 Skids (2)
- 10 Core rail, locking pins and pigtail (Core)
- 11 Arm tubes (4), one shown
- 12 Arm root fittings (4), one shown
- 13 Motor mounts (4), one shown
- 14 Motors (8), two shown
- 15 Motor controllers (8), two shown
- 16 Pivot bolt and lock pin (4 each), one shown
- 17 Folding propellers (8), two shown
- 18 Battery deck
- 19 Pack guides (2)
- 20 GNSS mast
- 21 ColdCell packs (2) and straps (4)
- 22 Float release payload
- 23 Tether module payload



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-Lift

Figure 1. Every component in build order. One of the four arms is shown pulled apart; the others are the same.

Kitewright Lift is a coaxial eight-motor aircraft: a two-plate carbon hub with the Kitewright Core hung under it and its lid standing up inside, four carbon arms that fold down on machined aluminium hinges, a motor above and below each arm tip, a raised battery deck with two ColdCell packs on top, skid landing gear, and the Core's payload rail underneath. Two payloads are built with it: a line-and-float release and a tether power module with its ground set. Of the 39 lines in the parts list, 16 are made: carbon sheet routed to shape, carbon tube cut and drilled, aluminium

fittings machined, two printed saddles and a sewn sling. The rest are bought, and the Core and the packs are built to their own designs. The parts are estimated at USD 6,627 for the aircraft and USD 3,976 for the payloads and ground equipment. Ready to fly it weighs about 21.8 kg, and it is rated to carry 3.2 kg inside its 25 kg take-off limit.

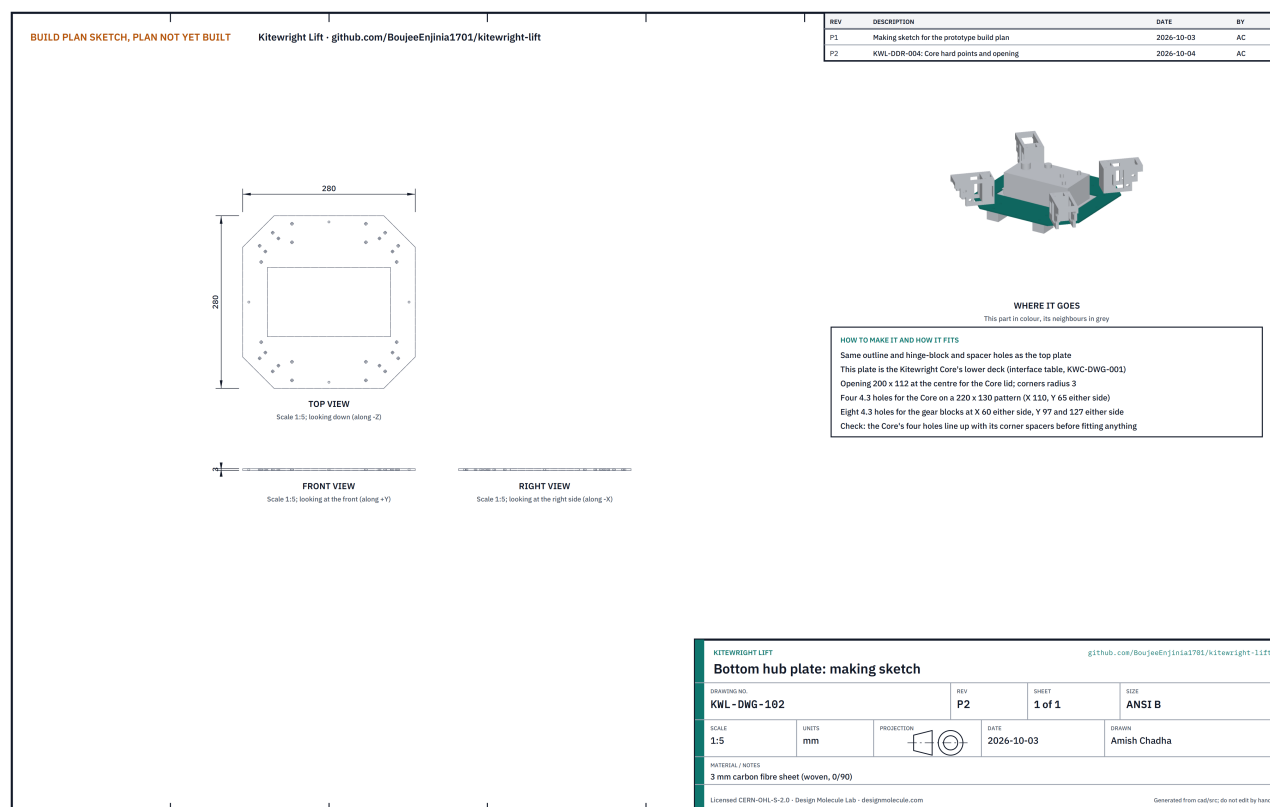
■ 2. What changed to make it buildable

Component	The concept had	The buildable design has	Why
Arm fold	Folding arms, no joint drawn	A machined hinge block at each hub corner: a slot for the arm's tongue, an 8 mm pivot bolt near the bottom, an 8 mm lock pin, and a stop bridge across the top	Lift presses the arm up against the bridge, so the arm cannot fold under power; the pin only stops it drooping
Arm root	Tube into the body	A one-piece aluminium fitting: a tongue in the hinge and a collar round the tube	A solid seat for the pivot and lock holes; the tube is bonded and bolted in
Hinge position	Not set	Pivot 180 mm from the centre and the plate corners cut at 162 mm	The folded arm otherwise cut into the bottom plate; now 6 mm clear
Central body	A body	Two 3 mm carbon plates 60 mm apart on the hinge blocks and four spacers; the Kitewright Core hung under the bottom plate, its lid up through a 200 x 112 mm opening	A light, stiff box, and the Core fitted to the family envelope that Range shares
Battery bay	Two heated packs	A 2 mm carbon deck on four 60 mm standoffs, above the upper rotors, two guides and four cam straps round both packs	ColdCell's 410 mm packs clear of the hinge pins and outside the rotor discs
Motor mount	Not drawn	A split clamp on the arm tip with a motor on its top face and one under it	Thick enough above the tube for the motor screws
Motor controllers	Not placed	Strapped to the arm near the tip	They fold with the arm; only two power leads cross the hinge
Landing gear	Wide gear	Four carbon struts in machined blocks, skids 450 mm apart, hub 500 mm up	Folded motors clear the skids and the ground
Payload bay	Clear space	The Core's rail, two locking pins and DS-014 pigtail; each payload on a payload shoe to KWC-DWG-106	Every Kitewright payload fits Lift and Range the same way
Float release	A release	Payload shoe, printed saddles, a servo pin release and a sewn sling round a foam float	Bought release, simple made parts
Tether module	Converter and reel	Payload shoe, four posts, 4 kW converter, breakaway connector; ground supply, reel and safety set	Fits the same rail; parts if the tether snags
Fittings	Solid blocks	Pockets, windows and cut-away corners in each hinge block, side pockets and a window in each root fitting, and a 40 mm motor clamp	About a third of their first weight; 0.79 kg off the four arms

■ 3. Making the components

Sizes are in millimetres unless stated. Carbon sheet and tube: cut with a diamond or carbide tool, wear a dust mask and gloves, and seal cut edges with thin epoxy. Aluminium fittings are 6061-T6, machined by a machine shop or an online CNC service from the making sketches; deburr every edge.

3.1 Bottom hub plate



What it is and what it is made from. The floor of the hub: 3 mm woven carbon sheet, 280 mm square with each corner cut square to its arm.

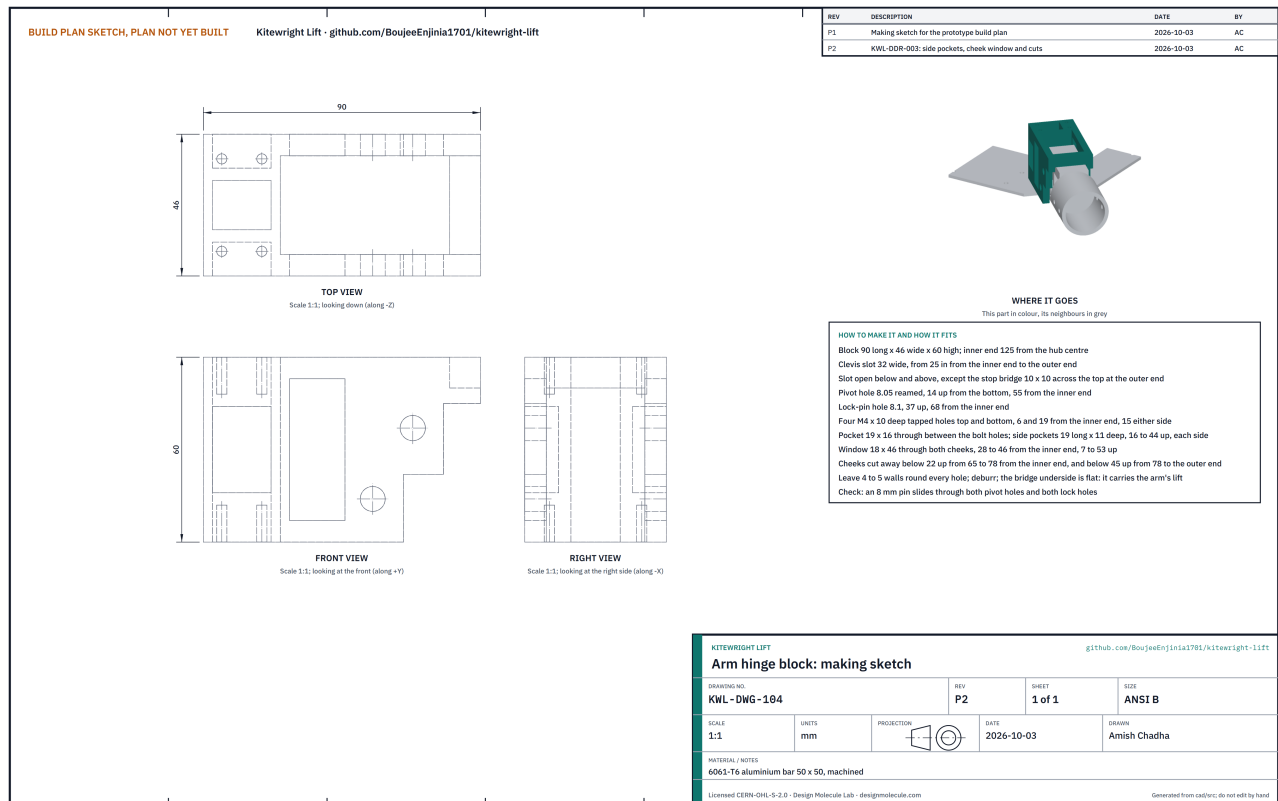
How to make it.

1. Route the outline: 280 x 280 with each corner cut off by a 72 mm face, 162 mm from the centre measured along the diagonal.
2. Drill four 4.3 mm holes per corner for the hinge block: on the diagonal, 131 and 144 mm out from the centre, 15 mm either side.
3. Drill four 3.3 mm spacer holes 130 mm from the centre on the forward, back, left and right lines.
4. Cut the 200 x 112 mm opening for the Core's lid at the centre, and drill the Core's four 4.3 mm holes on a 220 x 130 mm pattern round it. This plate is the Core's lower deck (the Kitewright interface table).
5. Drill eight 4.3 mm holes for the gear top blocks: 60 mm either side of the centre line fore and aft, 97 and 127 mm out to each side.
6. Seal the edges, the opening's included.

How it fits the parts next to it. The hinge blocks and spacers sit on its top face; the gear top blocks and the Core hang under it, the Core on its own four 8 mm corner spacers. Every fixing is a screw through the plate into a tapped hole or a nut.

Check before moving on. Lay the top plate on it: every shared hole lines up; the Core's corner spacers line up with its four holes.

3.2 Arm hinge blocks (four)



What it is and what it is made from. The joint between hub and arm, and the spacer between the hub plates at each corner: 6061-T6 aluminium from 50 x 50 bar, 90 x 46 x 60 mm.

How to make it.

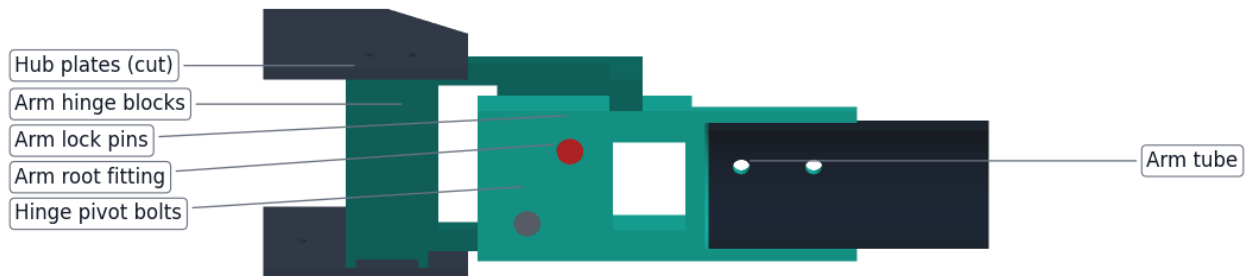
- Square the block to 90 long, 46 wide, 60 high.
- Mill the 32 mm slot from 25 mm in from the inner end to the outer end, open at the bottom, and open at the top except for a 10 x 10 mm bridge across the outer end.
- Ream the 8.05 mm pivot hole through both cheeks, 14 mm up from the bottom and 55 mm from the inner end.
- Drill the 8.1 mm lock-pin hole, 37 mm up and 68 mm from the inner end.
- Tap four M4 holes 10 deep in the top face and four in the bottom face, 6 and 19 mm from the inner end, 15 mm either side.
- Mill the 19 x 16 mm lightening pocket through between the tapped holes.
- Mill a side pocket 19 mm long and 11 mm deep into each side of the bolting end, 16 to 44 mm up, so the tapped holes keep a solid floor.

8. Mill a window 18 x 46 mm through both cheeks, 28 to 46 mm from the inner end and 7 to 53 mm up.
9. Cut the cheeks away below 22 mm up from 65 to 78 mm from the inner end, and below 45 mm up from 78 mm to the outer end. This leaves a 15 mm rail along the top of each cheek to carry the bridge.
10. Leave at least 4 to 5 mm of metal round every hole. Deburr. The underside of the bridge must be flat and smooth: it carries the arm's lift.

How it fits the parts next to it.

Arm hinge, cut open on the arm's centre line

Lift presses the tongue up against the stop bridge; the lock pin only stops the arm drooping



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

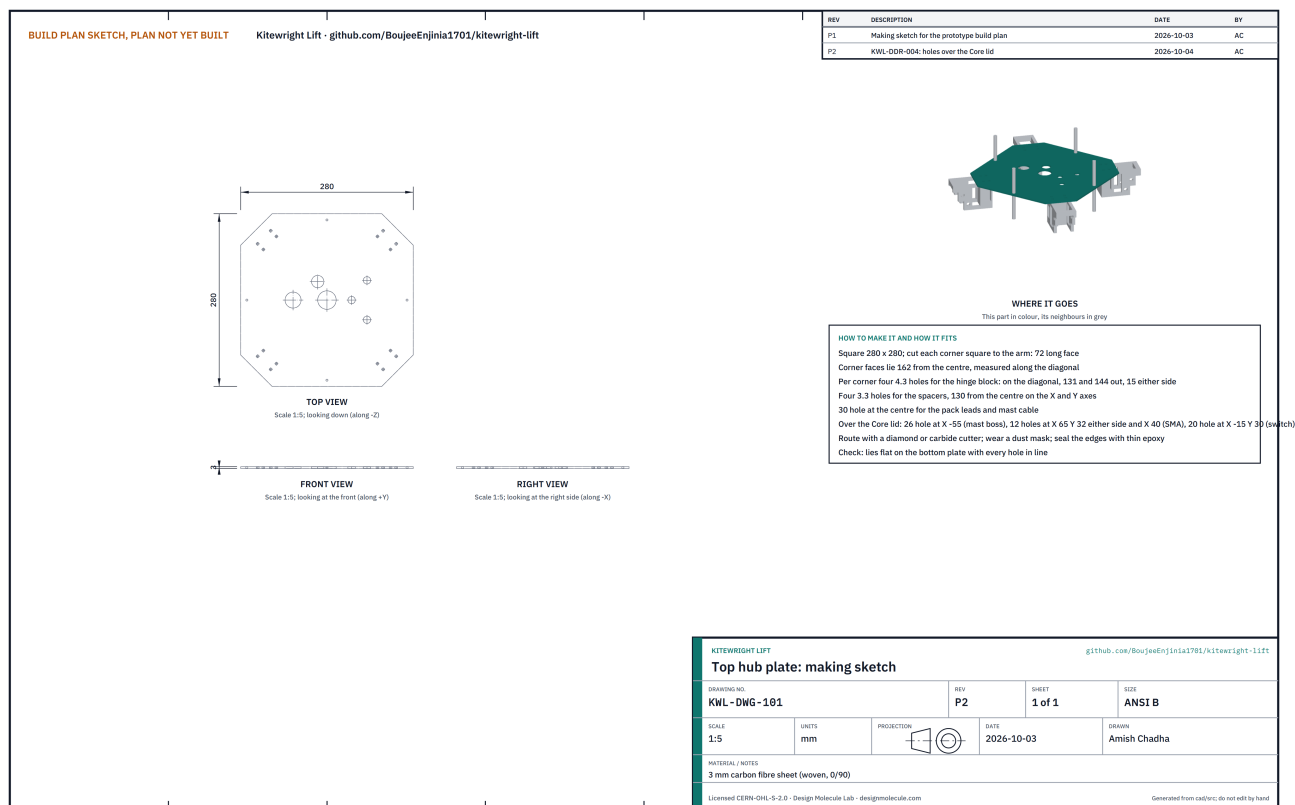
github.com/BoujeeEnjinia1701/kitewright-lift

Figure 2. The arm hinge cut open on the arm's centre line. In flight the tongue's top presses up on the bridge; the pivot bolt is near the bottom; the lock pin stops the arm drooping.

The block sits between the hub plates with its slot facing out along the diagonal. The arm's tongue fits the slot with 1 mm each side, turns on the pivot bolt, and in flight bears up against the bridge. The lock pin passes through both cheeks and the tongue.

Check before moving on. An 8 mm pin slides through both pivot holes and both lock holes in one line.

3.3 Top hub plate



What it is and what it is made from. The roof of the hub, 3 mm carbon sheet, the same outline as the bottom plate.

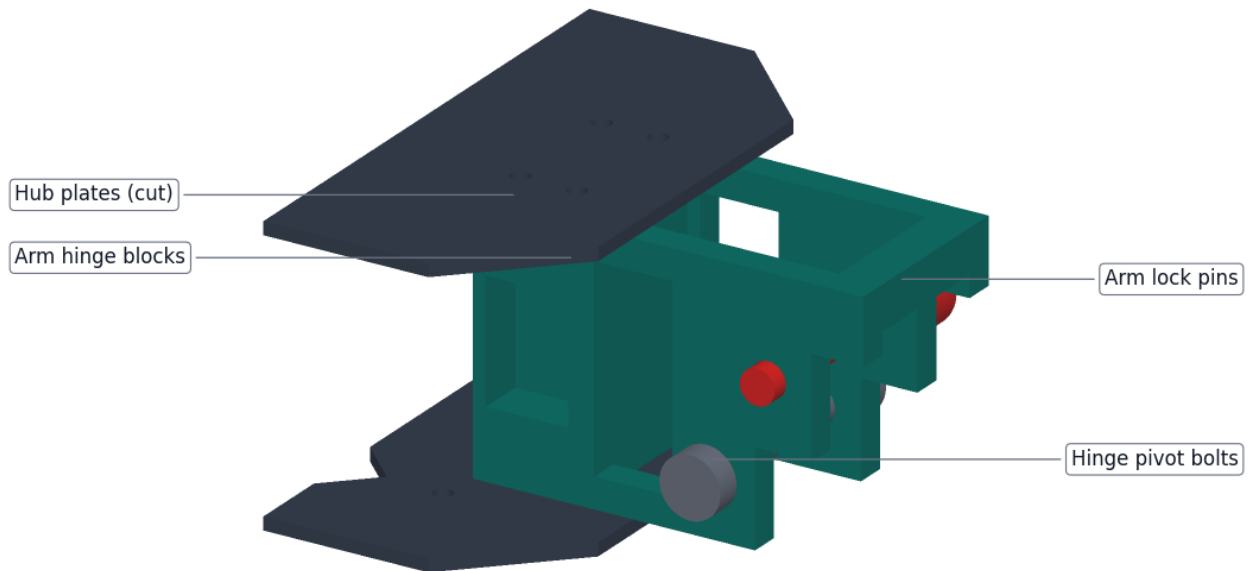
How to make it.

1. Route the same outline as the bottom plate.
2. Drill the same hinge-block and spacer holes.
3. Drill a 30 mm hole at the centre for the pack leads and the mast cable.
4. Over the Core's lid: a 26 mm hole 55 mm behind the centre for its GNSS mast boss, three 12 mm holes over its SMA bulkheads (65 mm ahead, 32 mm either side, and 40 mm ahead on the centre line) and a 20 mm hole over its safety switch (15 mm behind, 30 mm to the left).
5. Seal the edges.

How it fits the parts next to it.

Hinge block between the hub plates

Four M4 screws up through the bottom plate and four down through the top plate into each block



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 3. Each hinge block is held by four M4 screws up through the bottom plate and four down through the top plate.

Check before moving on. It lies flat on all four hinge blocks and spacers with no rocking.

3.4 Hub spacers and deck standoffs (bought)

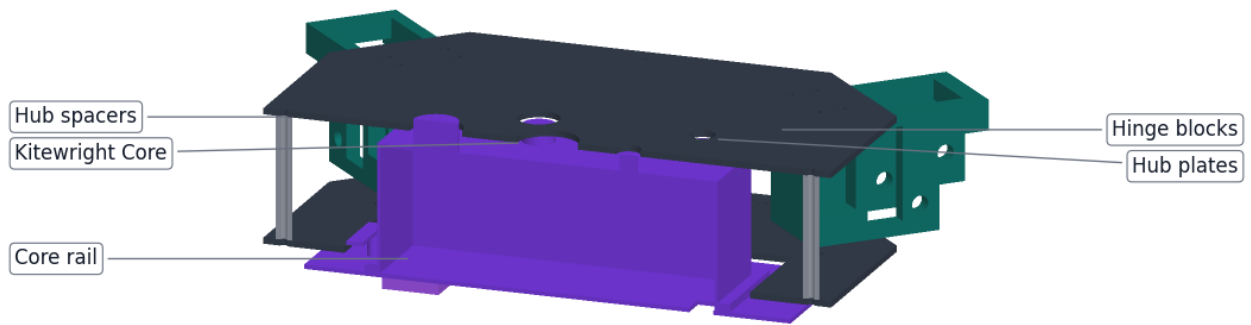
Buy eight 60 mm aluminium round standoffs, 8 mm across, M3 inside both ends. Four hold the hub plates apart beside the Core's lid; four carry the battery deck high enough that its underside clears the upper rotor blades by 9 mm. One long M3 screw through the deck, a standoff, the top plate and a spacer ties each stack.

3.5 Kitewright Core (built to the Core design)

Build the Kitewright Core, with its rail, two locking pins and DS-014 pigtail, to the Kitewright Core build plan (KWC-BLD-001). Lift takes it to the family mounting envelope (the Kitewright interface table, 2026-10-04): the Core plate hangs under the bottom hub plate on its four 8 mm corner spacers with four M4 screws on the 220 x 130 mm pattern, and its lid stands 55 mm up through the 200 x 112 mm opening into the hub, its mast boss, SMA bulkheads and switch reaching up into holes in the top plate. Lift has no room for the Core's own antennas and mast above the lid: the Core's GNSS receiver goes on Lift's deck mast (section 3.16) and its three antennas on 300 mm SMA extension leads clipped to the gear struts, pointing down. The Core comes with solder pads and a strain-relief bar but no power leads; Lift supplies them (section 3.17a). It weighs 0.99 kg.

Kitewright Core under the hub, cut open

Core plate on four 8 mm spacers under the bottom plate; its lid stands up through the opening into the hub

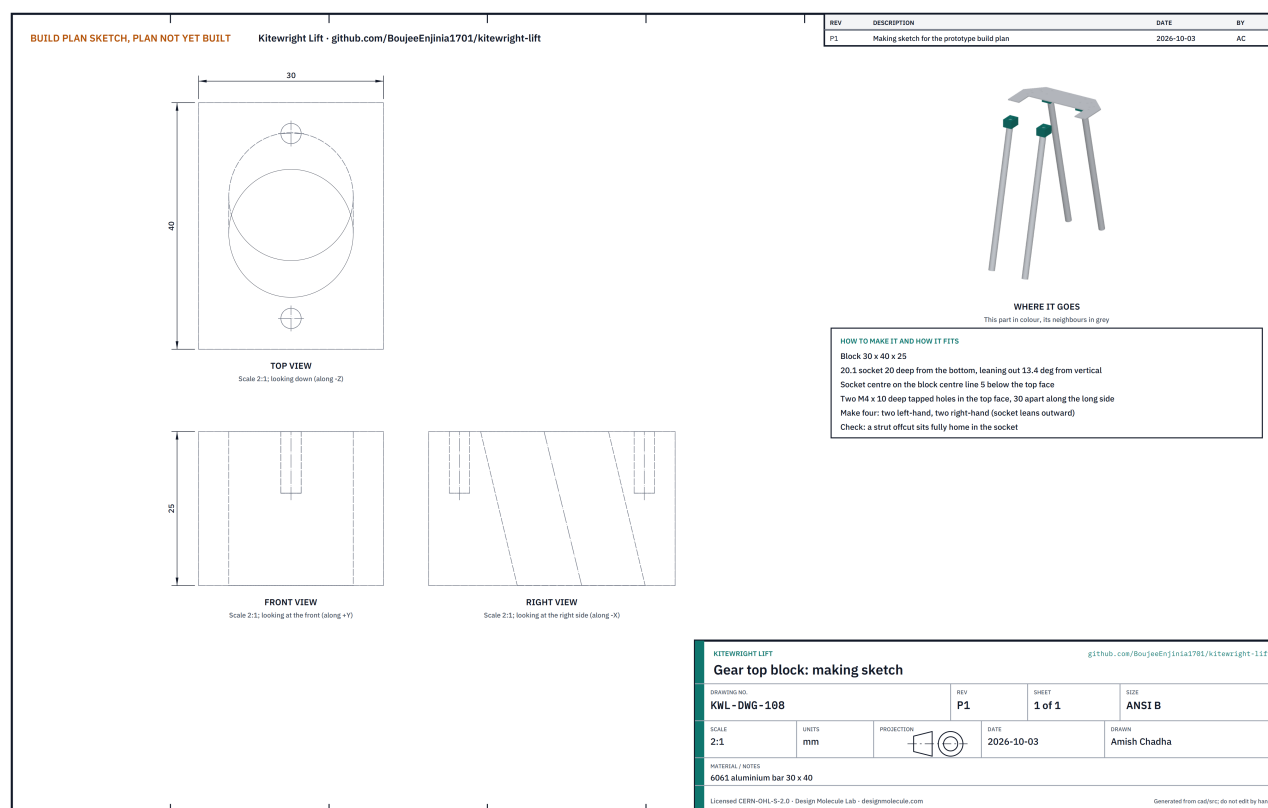


BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 4. The Core plate on its 8 mm spacers under the bottom plate, its lid standing up through the opening into the hub between the spacers and hinge blocks.

3.6 Gear top blocks (four)



What it is and what it is made from. The fixing that holds each gear strut under the hub: 6061 aluminium, 30 x 40 x 25 mm.

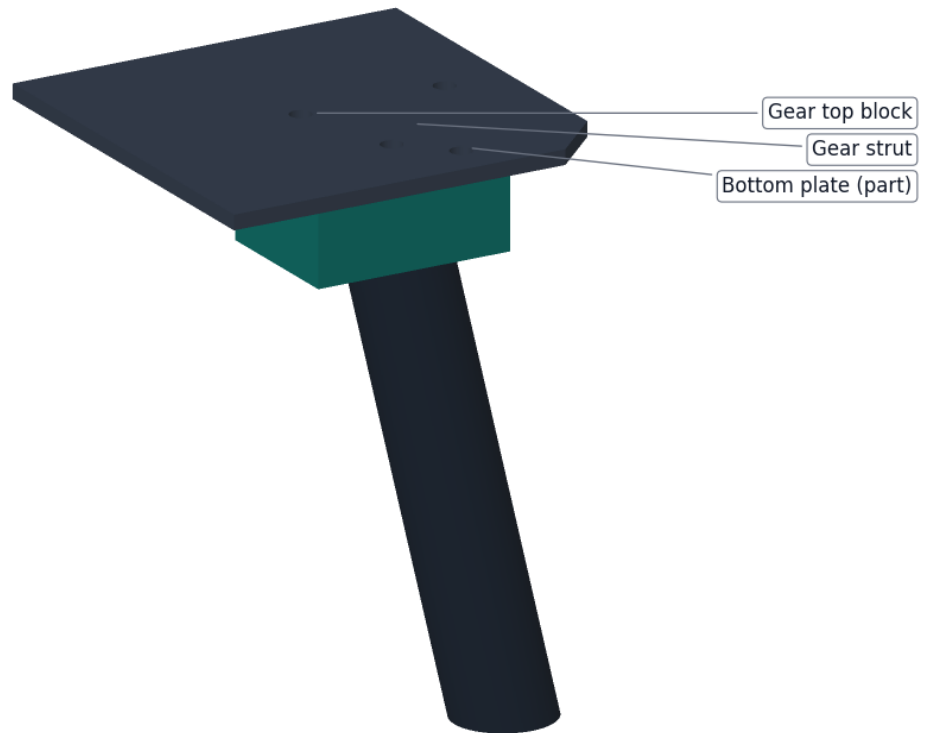
How to make it.

1. Square the block.
2. Bore a 20.1 mm socket 20 mm deep from the bottom, leaning outward 13.4 degrees from vertical, its centre on the block's centre line 5 mm below the top face.
3. Tap two M4 holes 10 mm deep in the top face, 30 mm apart.
4. Make two left-hand and two right-hand, so each socket leans outward.

How it fits the parts next to it.

Gear top block under the bottom plate, cut open

Two M4 screws down through the plate; strut bonded 20 mm into the angled socket



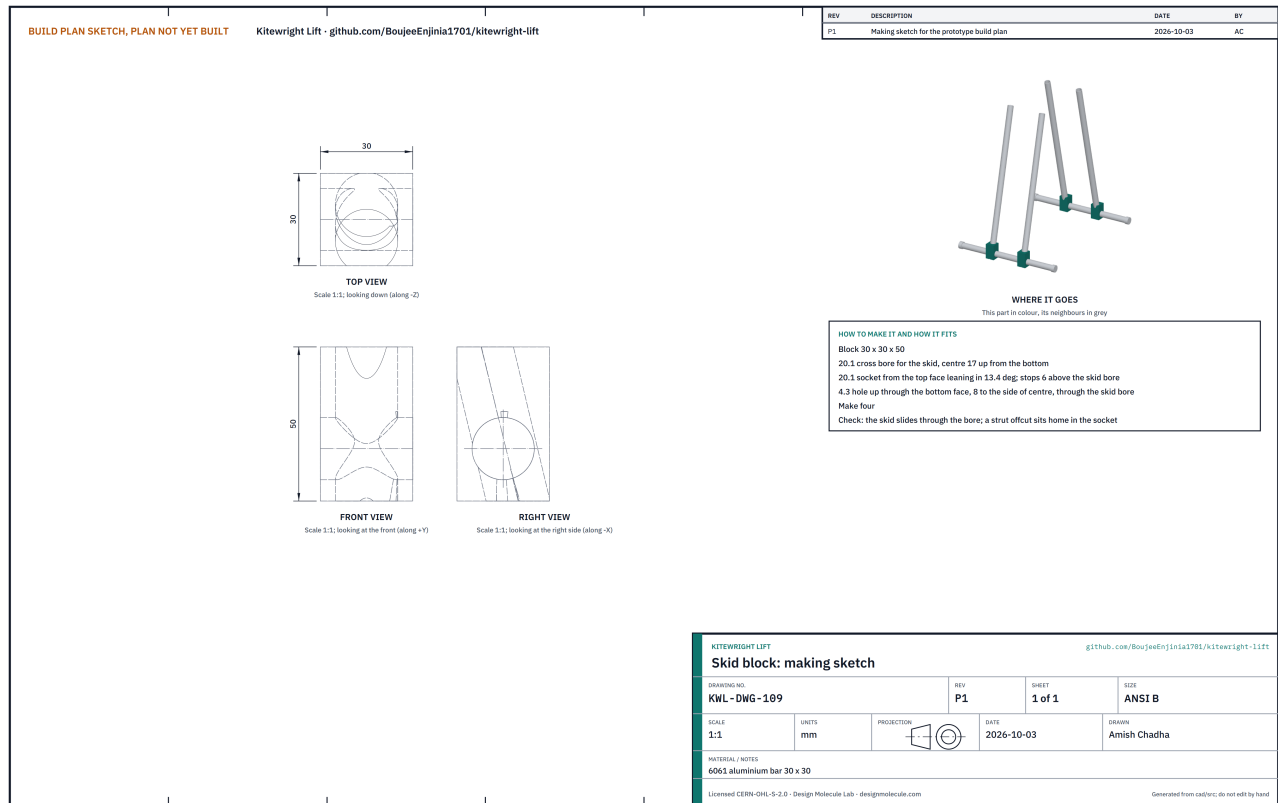
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 5. Two M4 screws down through the bottom plate hold the block; the strut is bonded into its socket.

Check before moving on. A strut offcut sits fully home in the socket.

3.7 Skid blocks (four)



What it is and what it is made from. The foot of each strut, where it meets the skid: 6061 aluminium, 30 x 30 x 50 mm.

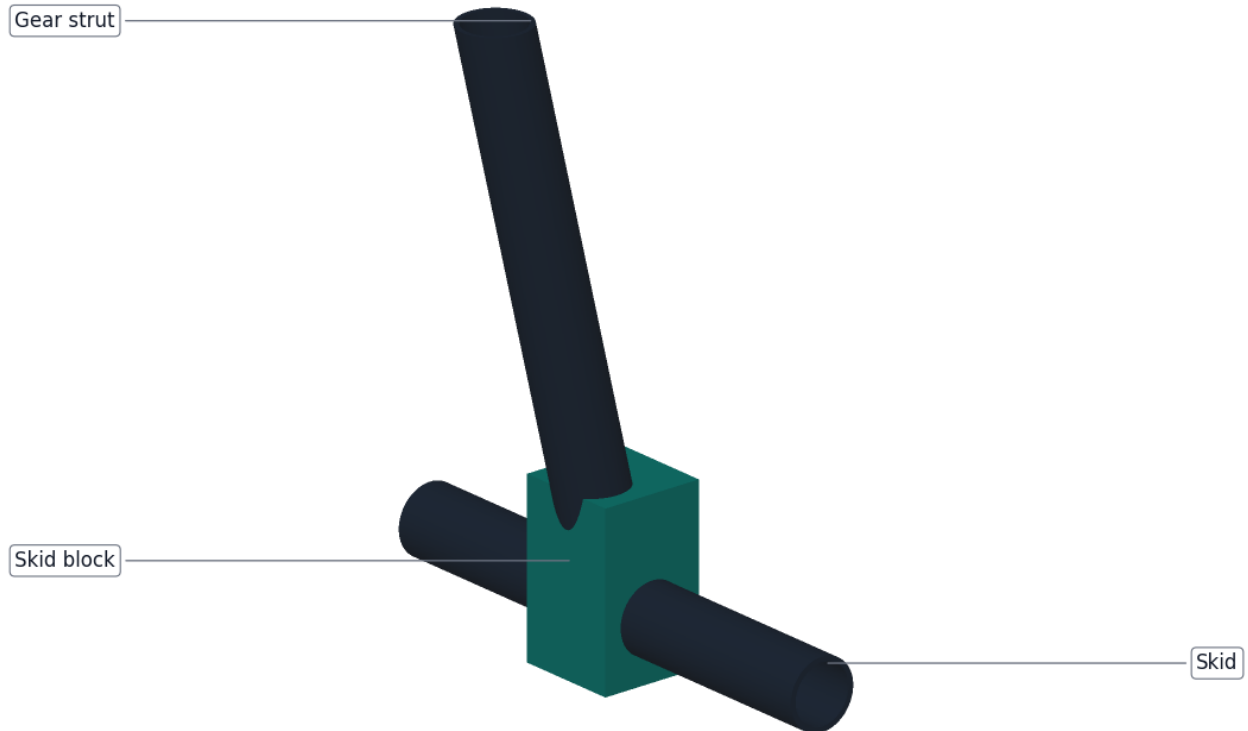
How to make it.

1. Square the block.
2. Bore the 20.1 mm cross bore for the skid, its centre 17 mm up from the bottom.
3. Bore the 20.1 mm strut socket down from the top face, leaning inward 13.4 degrees, stopping 6 mm above the skid bore.
4. Drill a 4.3 mm hole up through the bottom face, 8 mm to the side of centre, through the skid bore.

How it fits the parts next to it.

Skid block, strut and skid

Skid through the cross bore with an M4 bolt; strut bonded into the socket above it



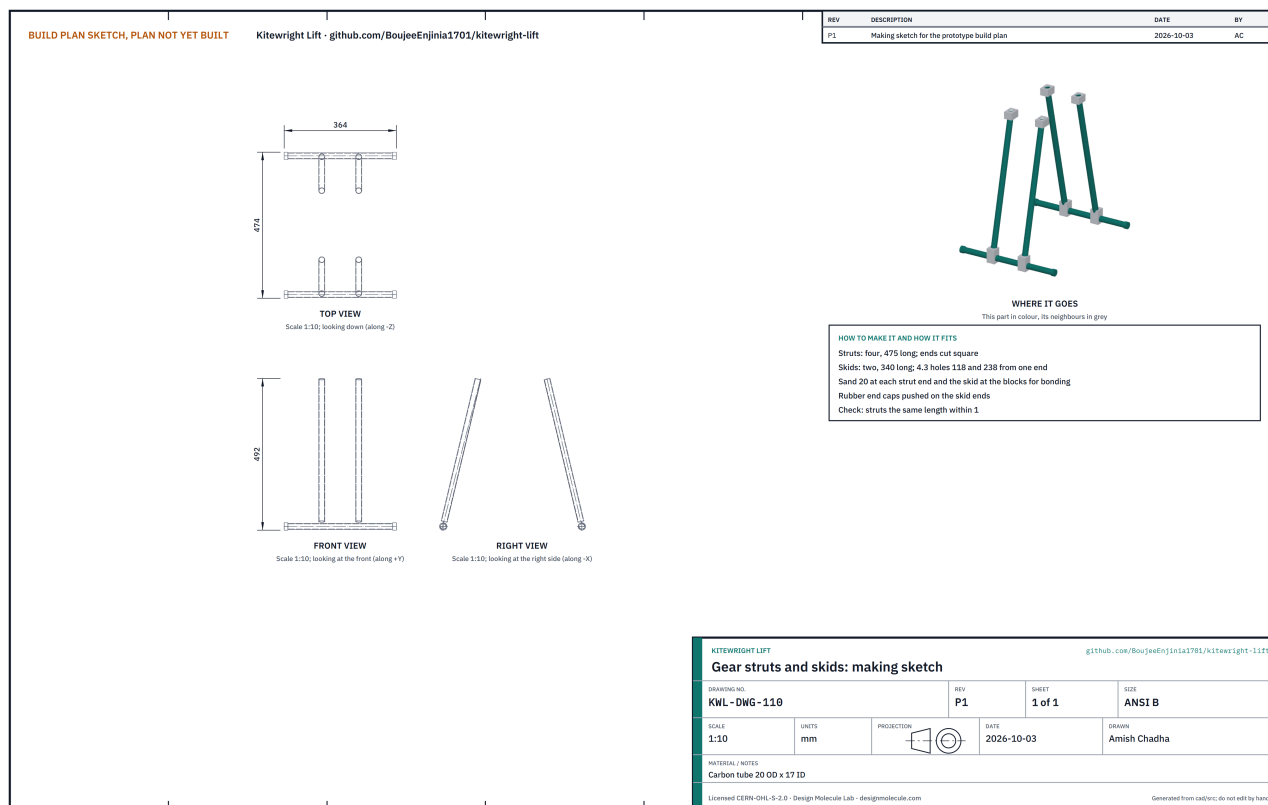
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 6. The skid passes through the cross bore and is held by an M4 bolt; the strut is bonded into the socket above it.

Check before moving on. The skid slides through the bore and a strut offcut sits home in the socket.

3.8 Gear struts and skids



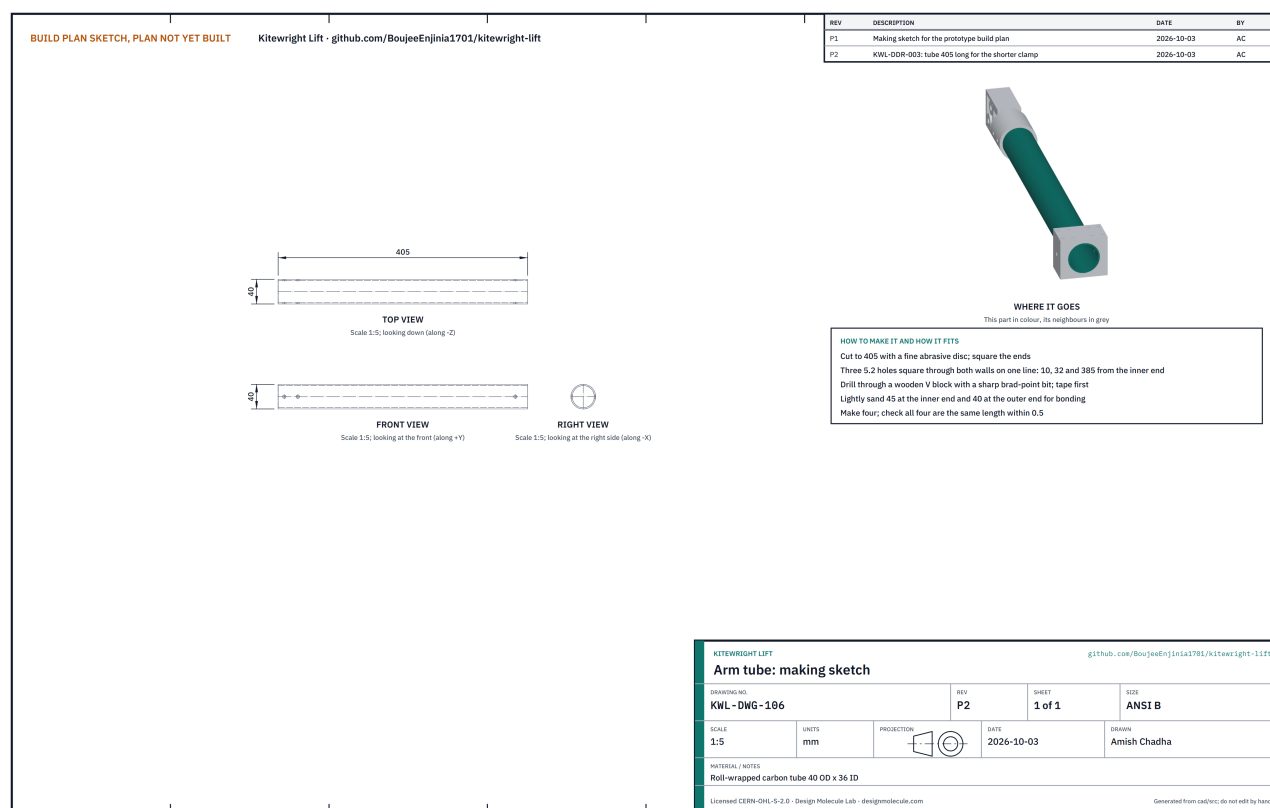
What it is and what it is made from. Carbon tube 20 mm across, 17 mm inside.

How to make it.

1. Cut four struts 475 mm long and two skids 340 mm long; square the ends.
2. Drill each skid with two 4.3 mm holes, 118 and 238 mm from the same end.
3. Sand the bonding areas: 20 mm at each strut end, and the skid where it passes through each block.
4. Push a rubber cap on each skid end.

Check before moving on. The four struts are the same length within 1 mm.

3.9 Arm tubes (four)



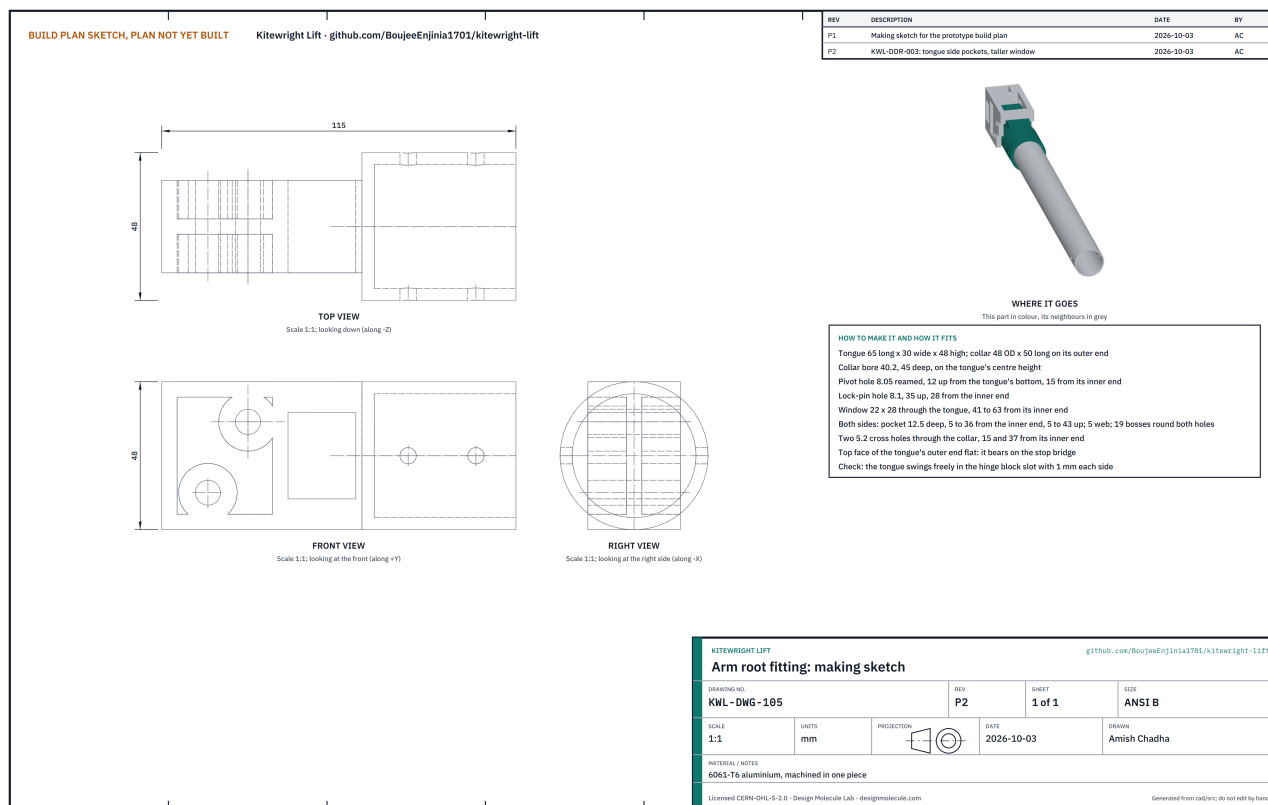
What it is and what it is made from. Roll-wrapped carbon tube 40 mm across, 36 mm inside, 405 mm long.

How to make it.

1. Wrap the cut line with tape and cut with a fine abrasive disc; square the ends.
2. In a wooden V block, drill three 5.2 mm holes square through both walls on one line: 10, 32 and 385 mm from the inner end.
3. Sand 45 mm at the inner end and 40 mm at the outer end for bonding.

Check before moving on. The four tubes are the same length within 0.5 mm.

3.10 Arm root fittings (four)



What it is and what it is made from. One machined 6061-T6 piece: a 65 x 30 x 48 mm tongue with a 48 mm collar on its outer end.

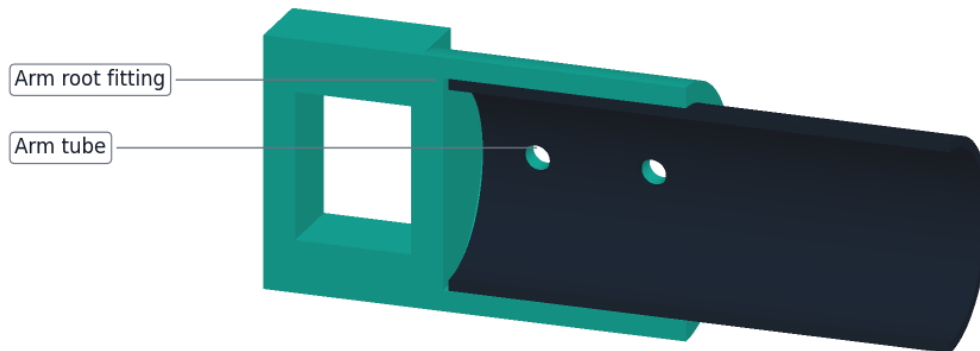
How to make it.

1. Turn the collar to 48 mm and bore it 40.2 mm, 45 mm deep, on the tongue's centre height.
2. Mill the tongue to 30 mm wide and 48 mm high.
3. Ream the pivot hole 8.05 mm, 12 mm up from the tongue's bottom and 15 mm from its inner end; drill the lock-pin hole 8.1 mm, 35 mm up and 28 mm from the inner end.
4. Mill the 22 x 28 mm window through the tongue, 41 to 63 mm from its inner end.
5. Mill a pocket 12.5 mm deep into each side of the tongue, 5 to 36 mm from its inner end and 5 to 43 mm up, leaving a 5 mm web in the middle and a 19 mm round boss of full width round the pivot hole and the lock-pin hole.
6. Drill two 5.2 mm cross holes through the collar, 15 and 37 mm from its inner end.
7. Keep the top face of the tongue's outer end flat: it bears on the bridge.

How it fits the parts next to it.

Arm tube in the root collar, cut open

Tube bonded 45 mm into the collar with structural epoxy and held by two M5 cross bolts



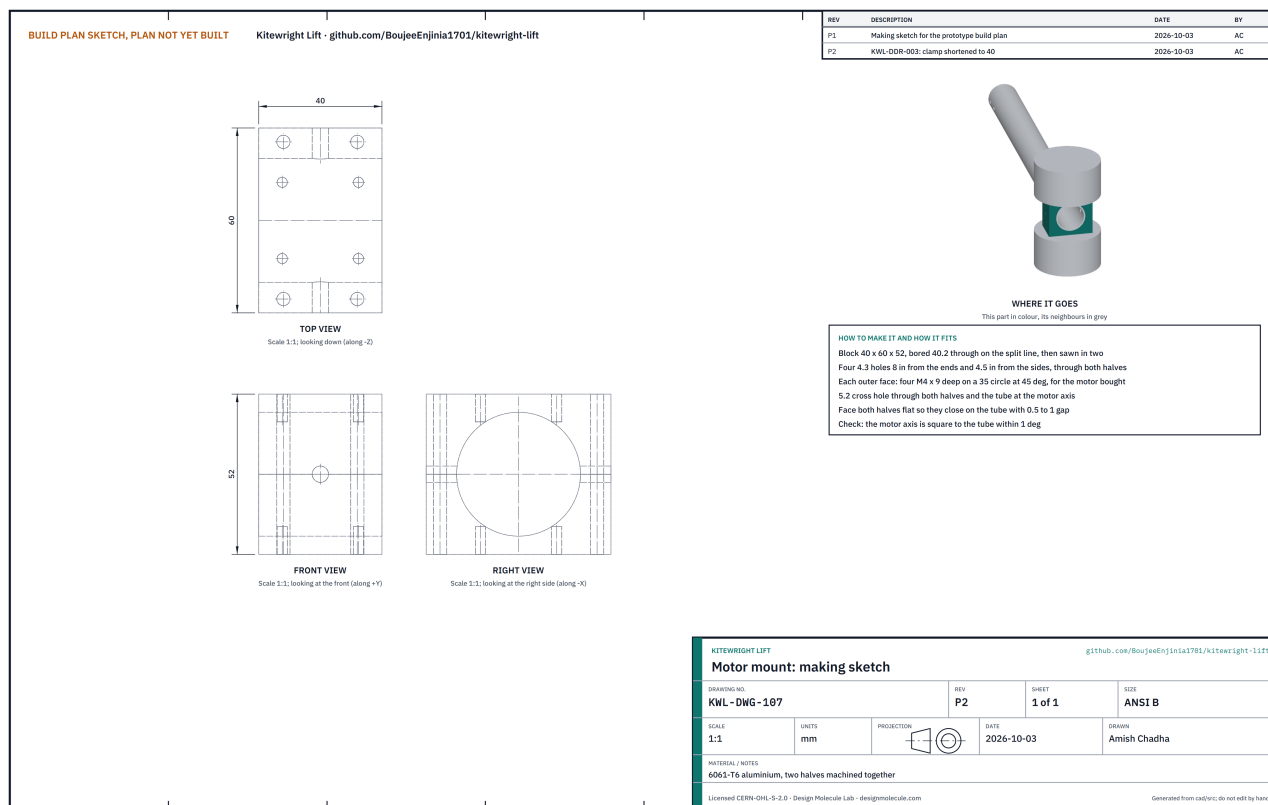
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 7. The tube goes 45 mm into the collar, bonded with structural epoxy and held by two M5 bolts through collar and tube.

Check before moving on. The tongue swings freely in a hinge block's slot with about 1 mm each side.

3.11 Motor mounts (four)



What it is and what it is made from. A split clamp on the arm tip, 40 x 60 x 52 mm (40 mm along the arm), two halves of 6061-T6 machined together.

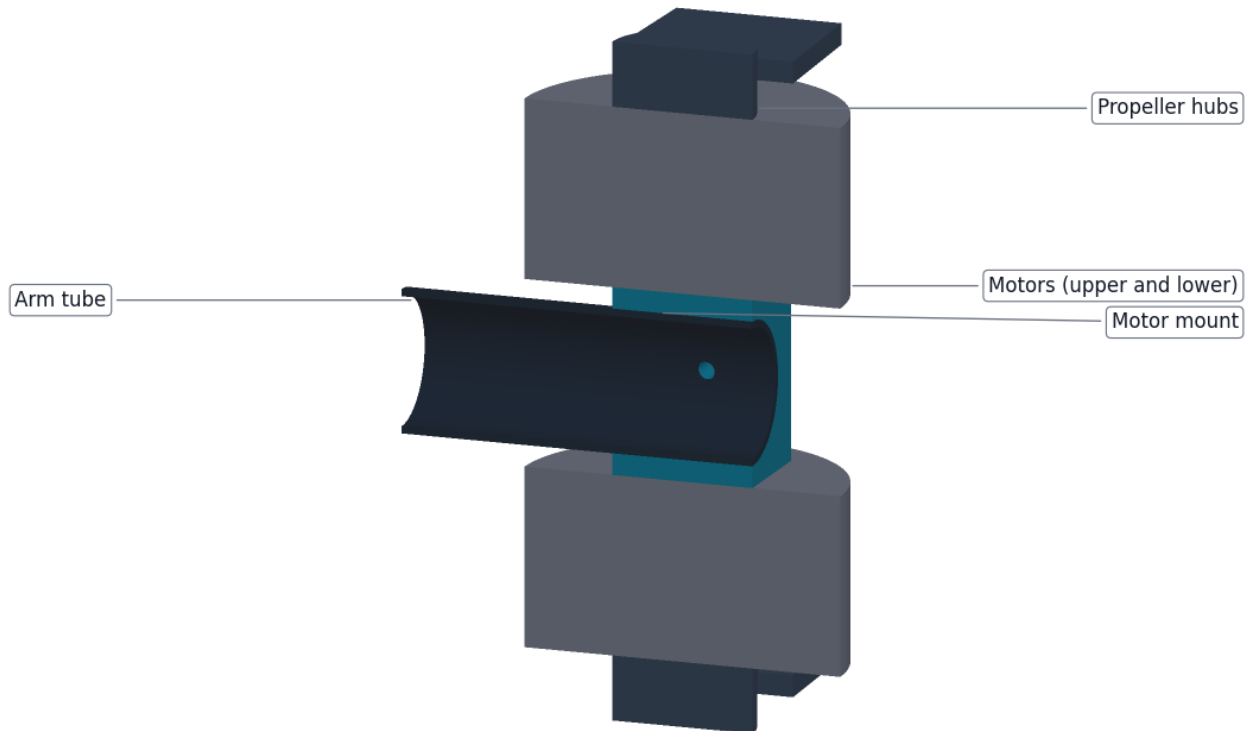
How to make it.

1. Square the block, bore it 40.2 mm on the split line, then saw it in two and face both halves so they close on the tube with a 0.5 to 1 mm gap.
2. Drill four 4.3 mm clamp holes through both halves, 8 mm in from the ends and 4.5 mm in from the sides.
3. Tap four M4 holes 9 mm deep in each outer face to suit the motors bought (a 35 mm circle on most motors of this class).
4. Drill a 5.2 mm cross hole through both halves at the motor axis.

How it fits the parts next to it.

Motor clamp at the arm tip, cut open

Two clamp halves on four M4 bolts; one M5 cross bolt through tube and clamp; a motor on each face



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 8. The clamp halves close on the tube with four M4 bolts and one M5 cross bolt through tube and clamp; the upper motor sits on the top face and the lower motor, upside down, on the bottom face.

Check before moving on. The motor axis is square to the tube within 1 degree.

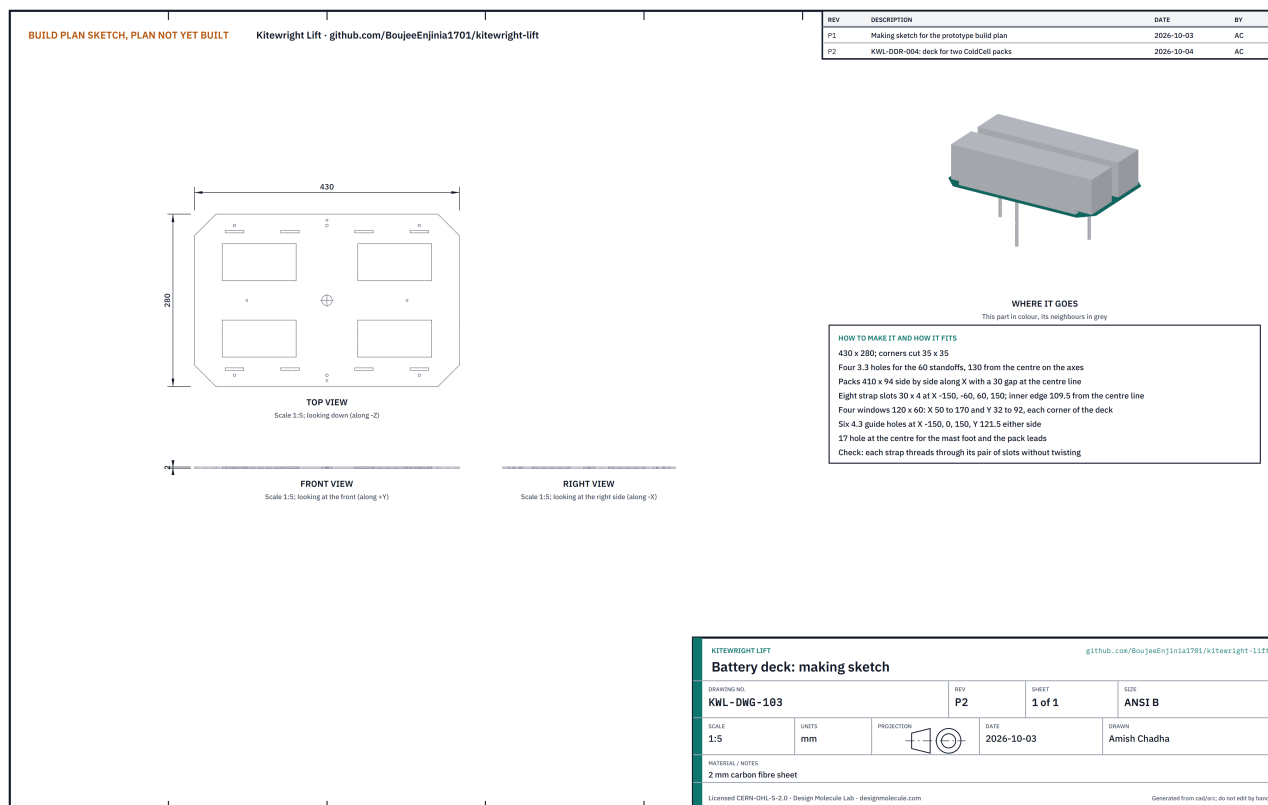
3.12 Motors, controllers and propellers (bought)

Buy eight brushless motors for 30 inch propellers, about 100 KV, rated 12S to 16S, about 450 g, with a maker's figure of about 10 kg static thrust on a 30 x 10 propeller at 51 V; eight controllers rated 16S and 80 A continuous with telemetry; and eight 30 x 10 carbon folding propellers, four turning each way. Upper and lower rotors on each arm turn opposite ways. Balance each propeller before fitting.

3.13 Pivot bolts and lock pins (bought)

Buy four stainless M8 shoulder bolts with an 8 mm shoulder 50 mm long, nyloc nuts and washers, and four 8 mm ball-lock pins with a grip of about 55 mm, a ring handle, a lanyard and a red flag. Tie each pin's lanyard to its hinge block so it cannot be lost.

3.14 Battery deck



What it is and what it is made from. The shelf the two ColdCell packs sit on: 2 mm carbon sheet, 430 x 280 mm with 35 mm corner cuts, sized for two 410 x 94 x 94 mm packs side by side with a 30 mm gap for the GNSS mast.

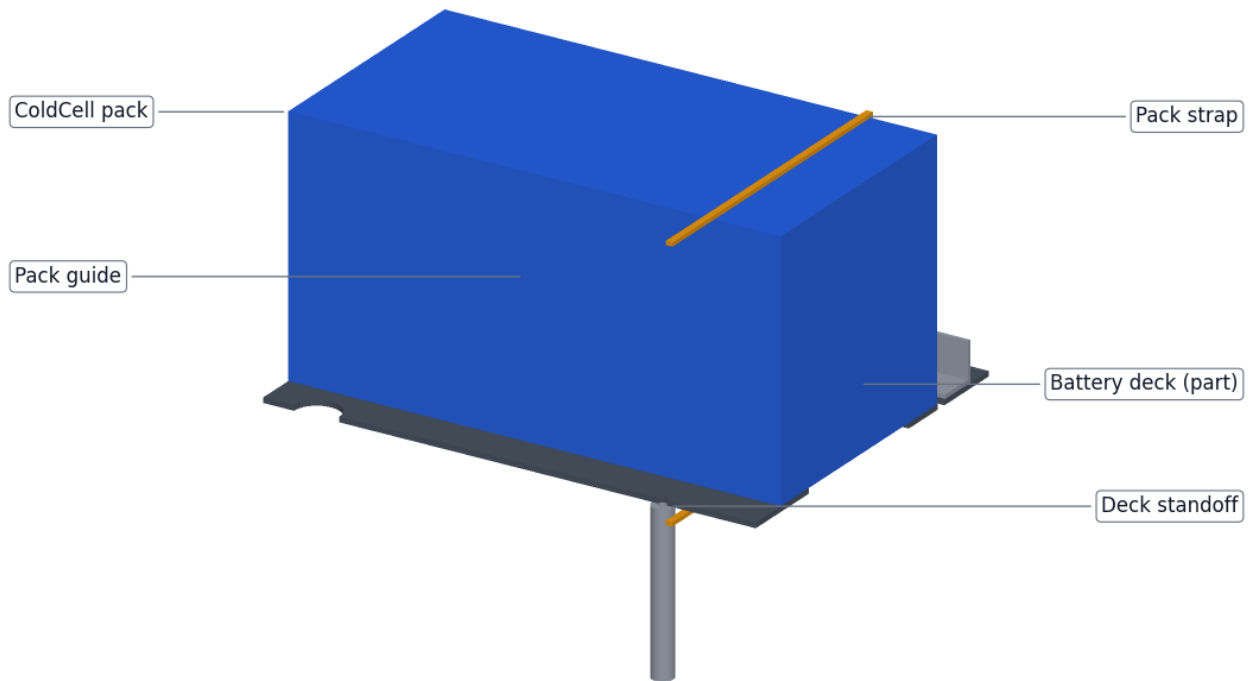
How to make it.

1. Route the outline.
2. Drill four 3.3 mm standoff holes 130 mm from the centre on the forward, back, left and right lines, and a 17 mm hole at the centre.
3. Cut eight 30 x 4 mm strap slots at 150 and 60 mm ahead of and behind the centre, their inner edges 109.5 mm out on each side.
4. Cut four 120 x 60 mm lightening windows, one under each pack end: 50 to 170 mm ahead of and behind the centre, 32 to 92 mm out on each side.
5. Drill six 4.3 mm guide holes at the centre and 150 mm either side, 121.5 mm out on each side.

How it fits the parts next to it.

Deck, pack guide, pack and strap

Packs side by side between the guides; each cam strap goes round both packs and through two deck slots



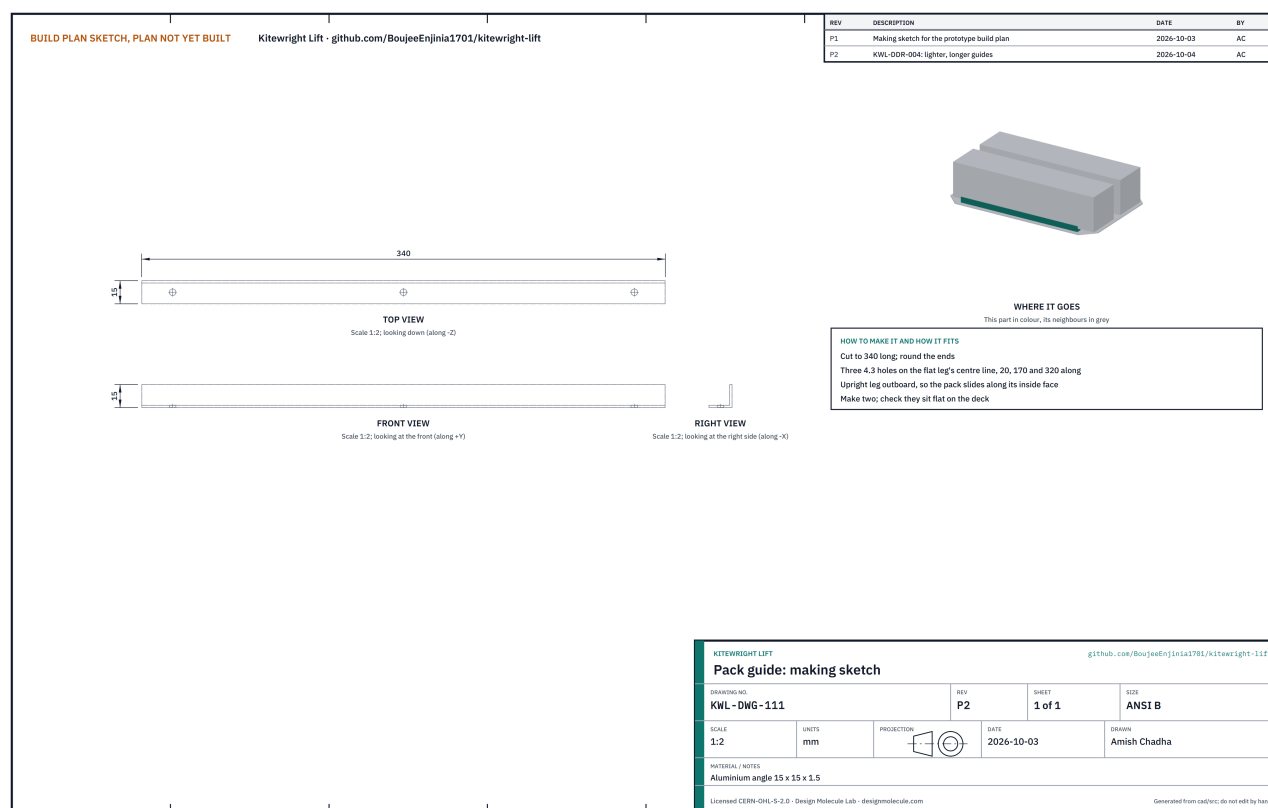
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 9. The packs sit side by side between the guides; each cam strap goes round both packs and through two deck slots.

Check before moving on. Each strap threads through its pair of slots without twisting.

3.15 Pack guides (two)



Cut two pieces of 15 x 15 x 1.5 mm aluminium angle 340 mm long, round the ends, and drill three 4.3 mm holes on the flat leg's centre line, 20, 170 and 320 mm along. Fit with the upright leg outboard. Check they sit flat on the deck.

3.16 Pack straps and GNSS mast (bought)

Buy four 25 mm cam-buckle straps 900 mm long with rubber backing (each goes round both packs), and a 16 mm carbon GNSS mast 230 mm long with a foot clamp; the Core's GNSS receiver goes on top of it.

3.17 ColdCell packs (built to the ColdCell design)

Build two lithium-ion ColdCell packs to the ColdCell build plan's variant for Lift: 14 cells in series and three in parallel, 21700 high-power lithium-ion cells of about 4.5 Ah rated for 45 A, with the external film heater, insulation, BMS and shell. Each pack is 50.4 V nominal (42.0 to 58.8 V), 680 Wh, 410 x 94 x 94 mm and 4.46 kg, with an AS150 power lead, exactly as ColdCell draws it (CCL-DWG-002).

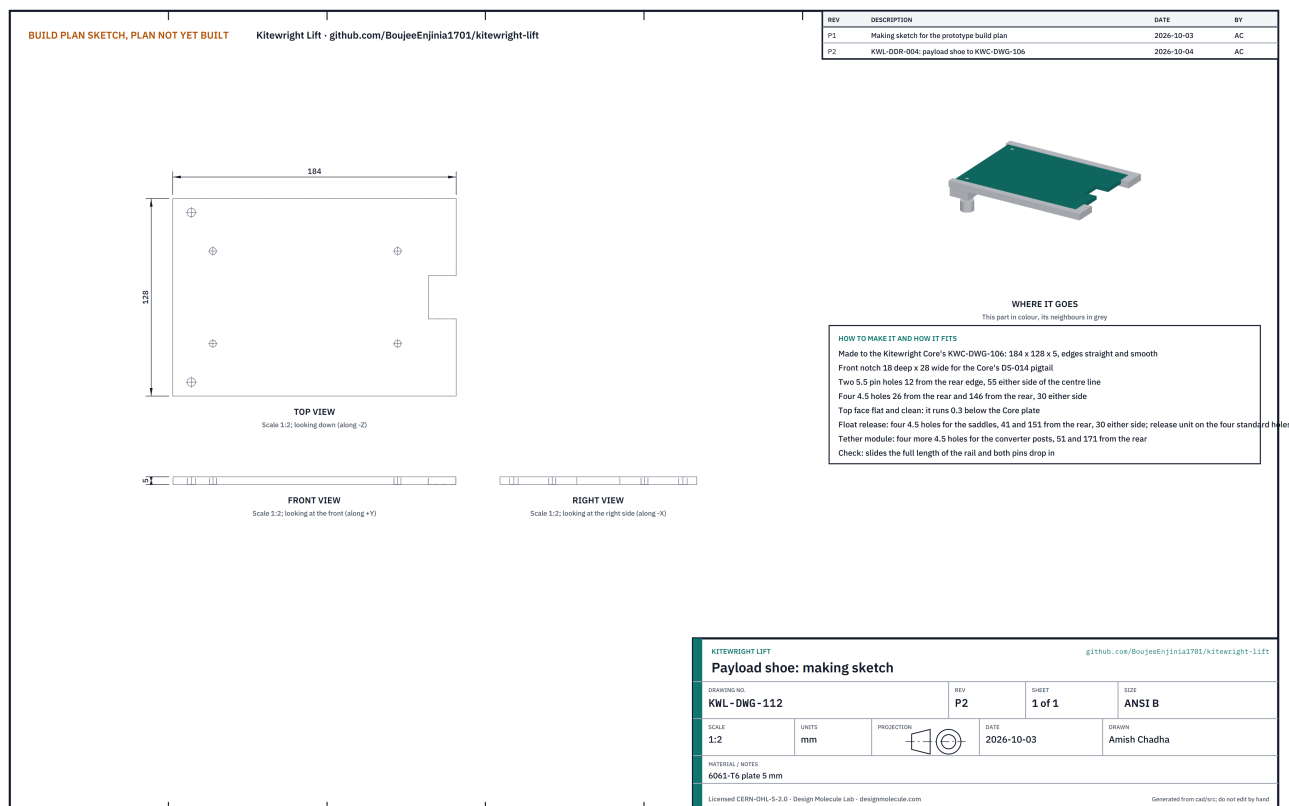
Operating note: warm weather. ColdCell keeps its 50 °C firmware warning for this pack until a TRL 4 measurement supports relaxing it. A full hover flight stays under the warning up to about 27 °C ambient. Above that, plan shorter flights: at 45 °C the warning comes after about 6 min of hover. Land when it comes.

Safety: Lithium-ion cells hold more fire energy than the LiFePO₄ cells first planned. Keep every ColdCell protection: the BMS blocks charging below 5 °C, the heater has cut-outs, every cell has its own fuse, the 50 °C warning lands the aircraft, and packs are charged only inside the fire-resistant charging box.

3.17a Power leads from the frame to the Core (bought, made up)

Lift supplies the power leads between its packs and the Core, and between the Core and its arms, because the Core no longer carries them. Make them from 8 AWG silicone wire, red and black, with four AS150 anti-spark connector halves: two for the pack inputs and two for the frame outputs. Solder the leads to the Core's pads, clamp them under its strain-relief bar and lead them out through the lid's grommets, keeping each lead as short as the route allows. Insulate every joint with heat-shrink before a pack is anywhere near.

3.18 Payload shoes (two)



What it is and what it is made from. The shoe every Kitwright payload is built on, made to the Core's drawing KWC-DWG-106: 6061-T6 plate, 184 x 128 x 5 mm.

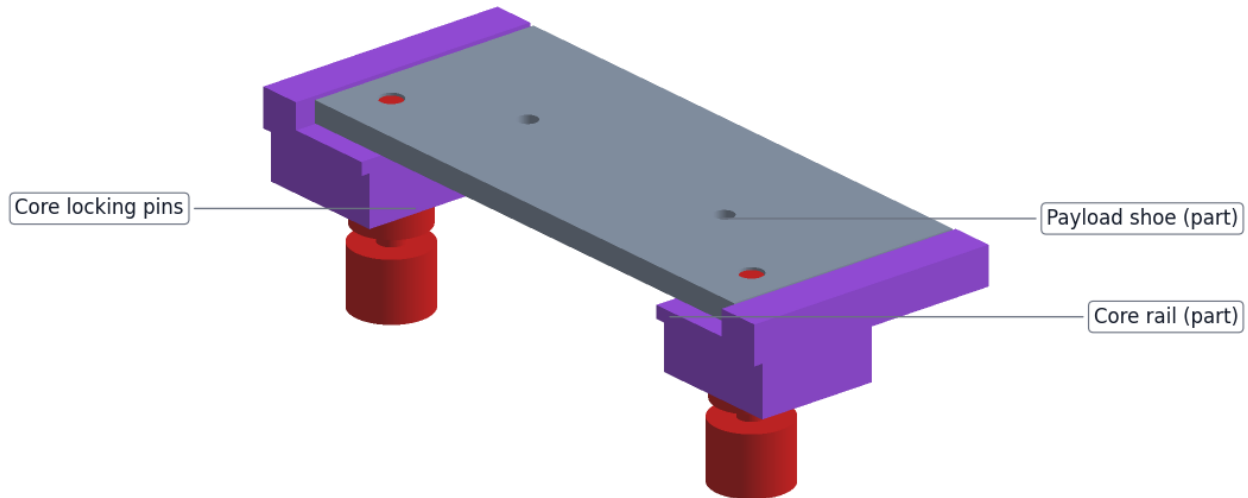
How to make it.

1. Cut the plate square with straight, smooth long edges: they ride on the Core rail's lips.
2. Cut the 18 x 28 mm notch in the front edge for the Core's DS-014 pigtail.
3. Drill the two 5.5 mm locking pin holes 12 mm from the rear edge, 55 mm either side of the centre line, and the four 4.5 mm holes 26 and 146 mm from the rear, 30 mm either side.
4. Float release shoe: four more 4.5 mm holes for the saddles, 41 and 151 mm from the rear, 30 mm either side.
Tether module shoe: four more 4.5 mm holes for the converter posts, 51 and 171 mm from the rear, 30 mm either side.

How it fits the parts next to it. Everything a payload hangs from the shoe stays inside 88 mm wide from the shoe down to 30 mm below the rail lips (the payload neck), clear of the Core's pin knobs at the back and its pigtail plug at the front.

Payload shoe on the Core rail, locked

The shoe's edges ride on the rail lips; both Core locking pins rise through the lips into the shoe



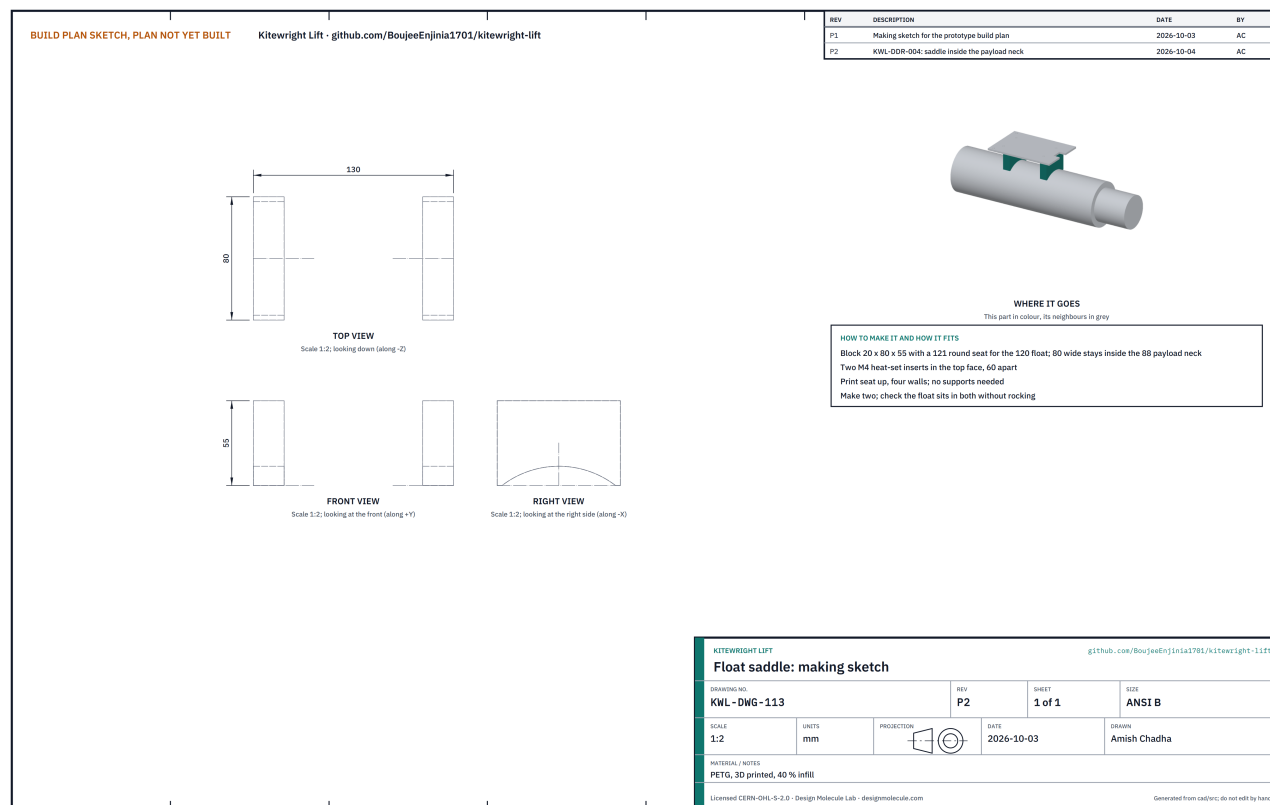
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 10. The shoe's edges ride on the rail lips under the Core plate; both locking pins rise through the lips into the shoe.

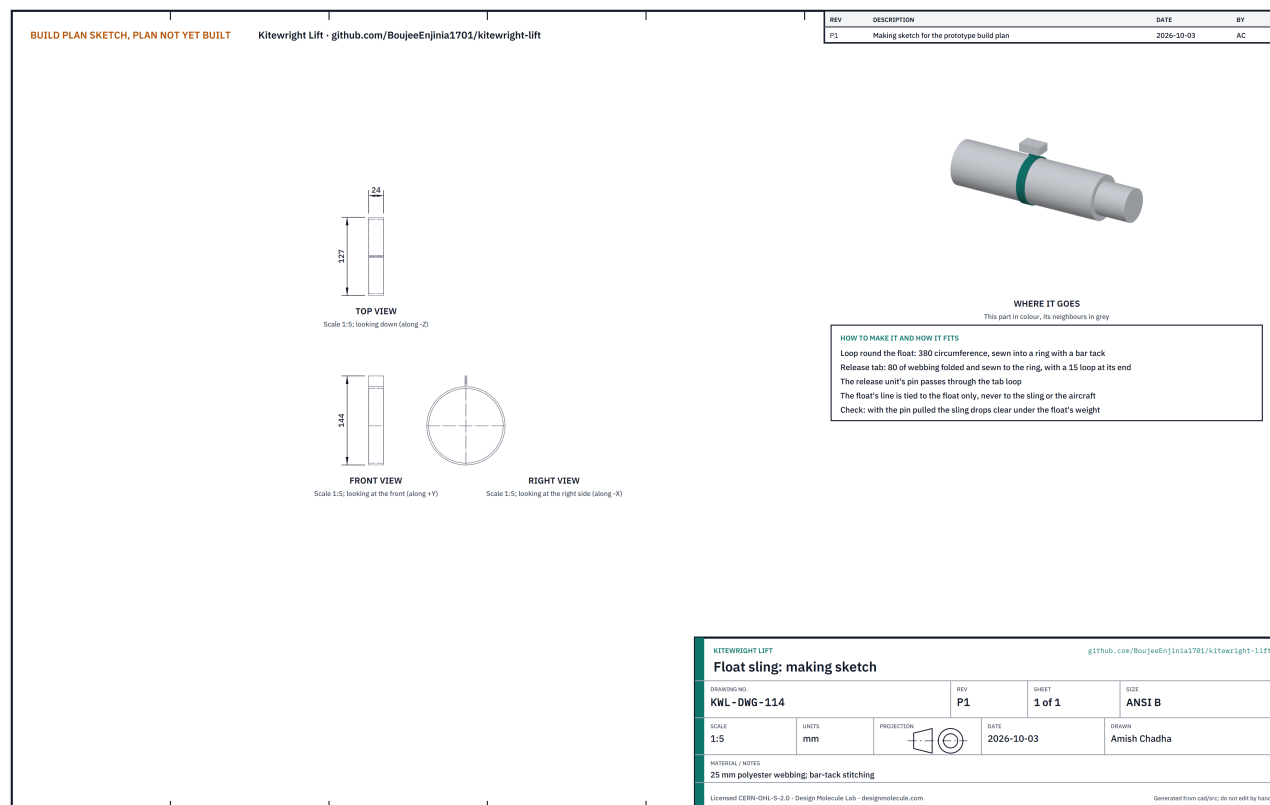
Check before moving on. The shoe slides the full length of the rail to the front stops and both pins drop in.

3.19 Float saddles (two)



Print two saddles in PETG, 20 x 80 x 55 mm with a 121 mm round seat, seat up, four walls and 40 % infill; set two M4 heat-set inserts in each top face 60 mm apart. At 80 mm wide they stay inside the payload neck, and at 55 mm tall the float hangs below it. Check the float sits in both without rocking.

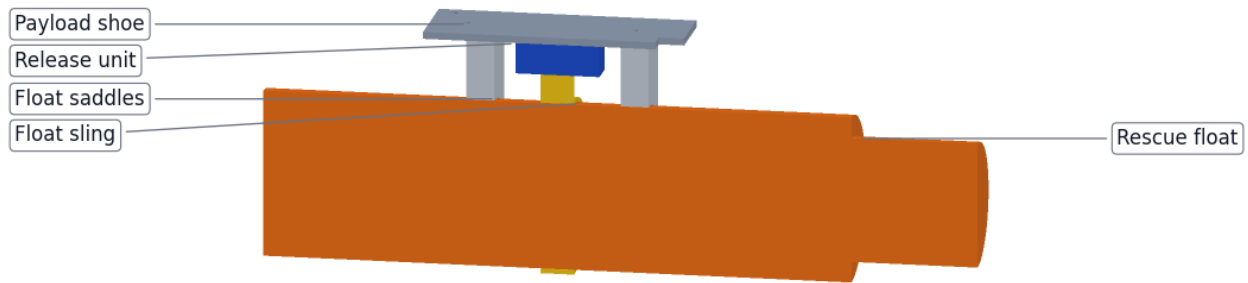
3.20 Float sling



Sew a ring of 25 mm polyester webbing 380 mm round to fit the float, with a bar tack. Sew an 80 mm tab to it with a 15 mm loop at its end for the release pin. The float's line is tied to the float only, never to the sling or the aircraft. Check that with the pin pulled the sling drops clear under the float's weight.

Float release, cut open

The release unit's pin holds the sling's tab; the float sits in two printed saddles under the shoe



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 11. The release unit's pin holds the sling's tab; the float sits in the two saddles.

3.21 Release unit, float and line (bought)

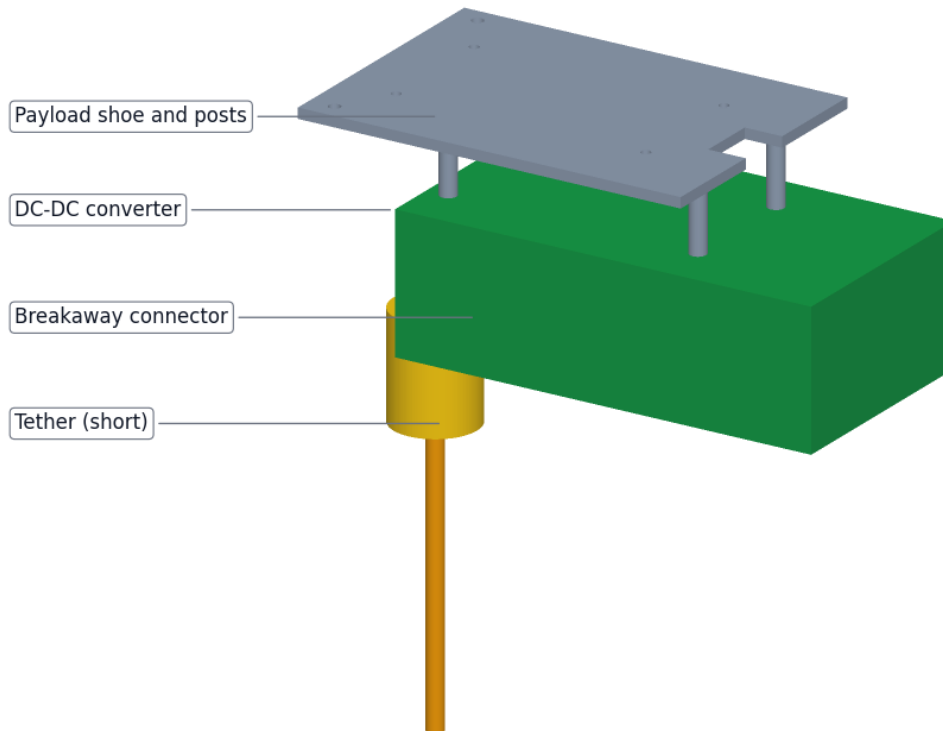
Buy a servo-driven pin release rated 10 kg that stays closed when unpowered, a closed-cell foam rescue float about 120 x 420 mm (about 0.6 kg), and 30 m of 6 mm floating line in a throw bag that clips to the float's end.

3.22 Tether module and ground set (bought)

Buy an isolated DC-DC converter (380 to 420 V in, 51 V out, 4 kW, conduction cooled, about 200 x 110 x 65 mm), a two-pole 600 V breakaway connector that pulls apart at about 200 N, 60 m of tether with two 0.75 mm² conductors and an aramid strength member, a 400 V DC isolated ground supply of 4.5 kW, a hand reel with a 600 V slip ring, and a ground safety set: insulation monitor, emergency stop, residual-current protection and an earth spike.

Tether module

Converter on four 34 mm posts under the shoe, below the payload neck; breakaway connector under its rear



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Figure 12. The converter on four 34 mm posts under the shoe, below the payload neck and 20 mm ahead of the Core's pin knobs; the breakaway connector under its rear end.

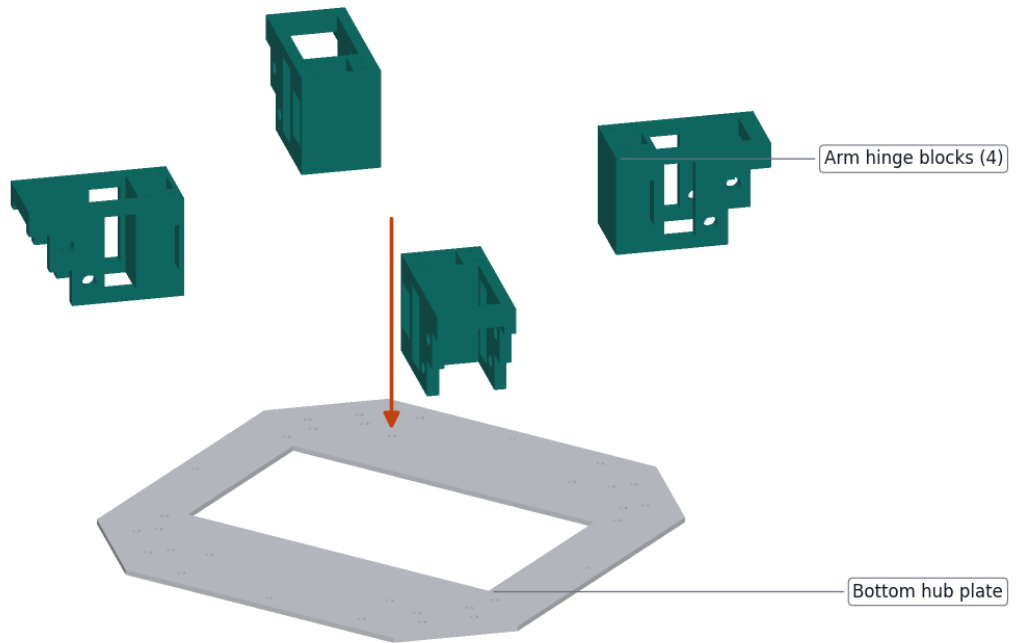
■ 4. Putting it together

Use medium threadlocker on every metal-to-metal screw. Bonded joints use structural epoxy on sanded, cleaned surfaces and cure for 24 h before load.

Step 1: Bolt the hinge blocks to the bottom plate

Step 1: Bolt the hinge blocks to the bottom plate

Four M4 x 10 screws up through the plate into each block; slot outward



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

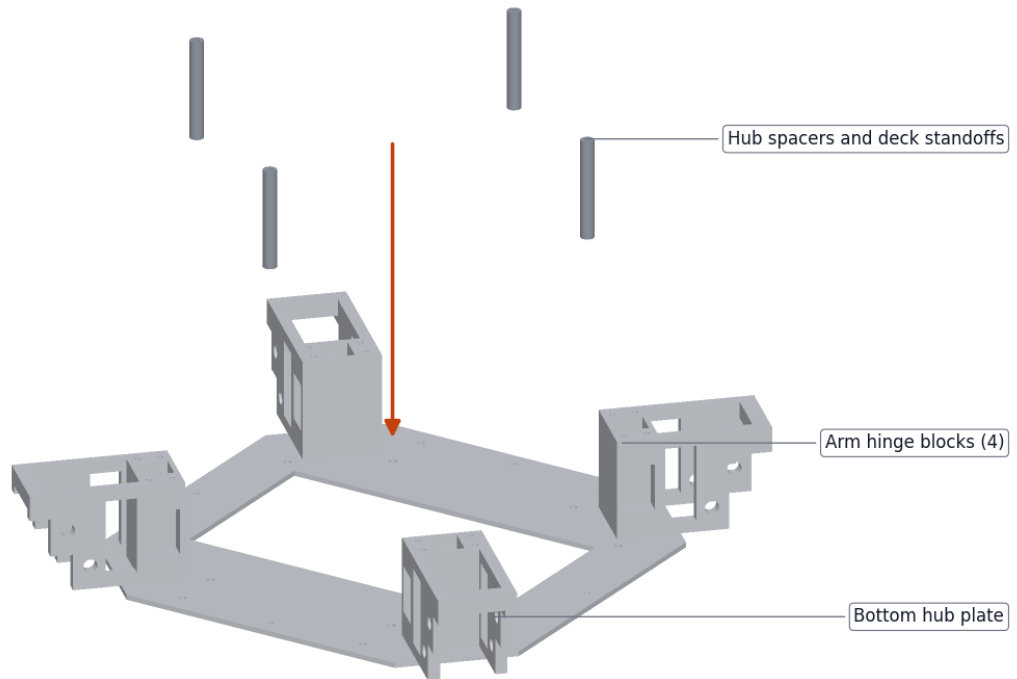
github.com/BoujeeEnjinia1701/kitewright-lift

Four M4 x 10 screws up through the plate into each block, slot facing out along the diagonal.

Step 2: Fit the hub spacers

Step 2: Fit the hub spacers

Four 60 mm spacers on M3 screws from below



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

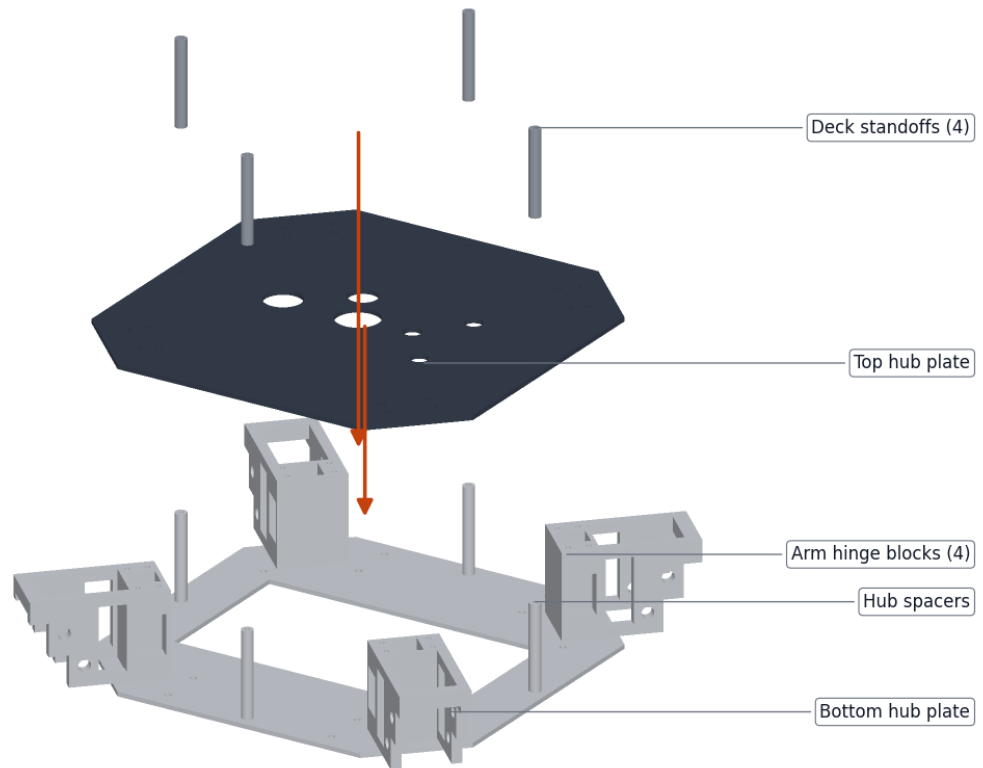
github.com/BoujeeEnjinia1701/kitewright-lift

The four 60 mm spacers on M3 screws from below.

Step 3: Fit the top hub plate

Step 3: Fit the top hub plate

Sixteen M4 screws into the blocks; 60 mm deck standoffs through it into the spacers



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

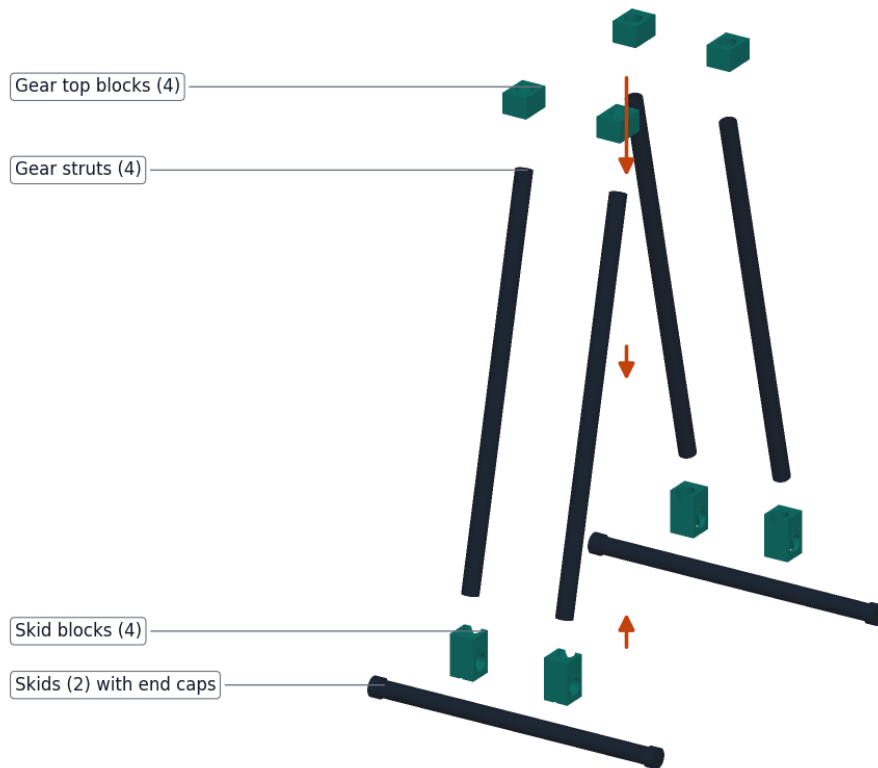
github.com/BoujeeEnjinia1701/kitewright-lift

Sixteen M4 screws down into the hinge blocks; the four 60 mm deck standoffs screwed through the plate into the spacers.

Step 4: Build the landing gear

Step 4: Build the landing gear

Bond the struts into the top and skid blocks on a flat board; skids through the bores, M4 bolts



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

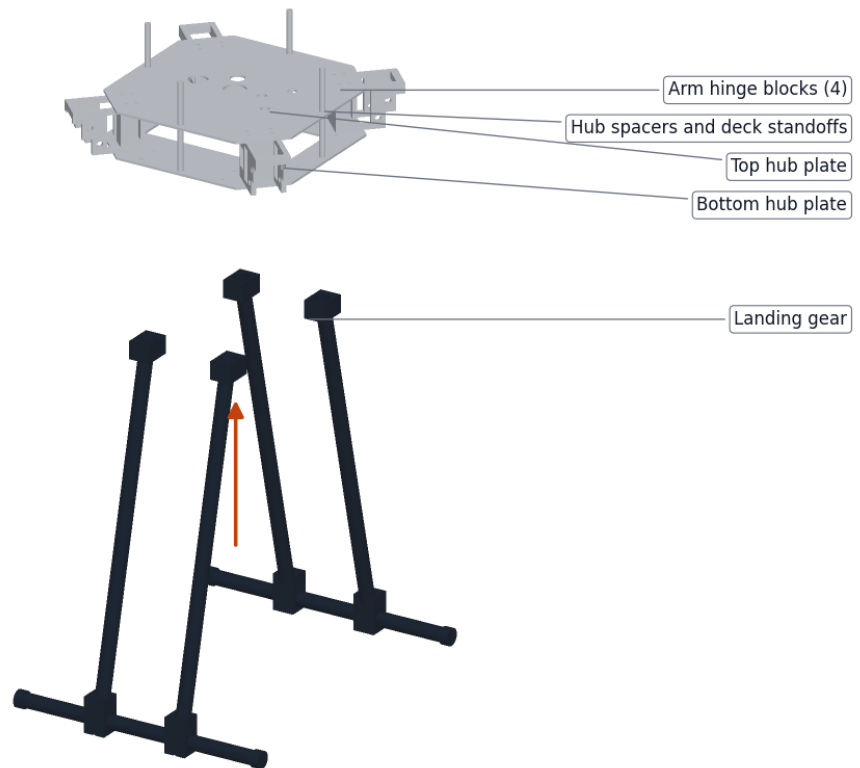
github.com/BoujeeEnjinia1701/kitewright-lift

On a flat board, bond the struts into the top and skid blocks with the skids through their bores, and bolt each skid with its M4 bolt. Cure 24 h.

Step 5: Bolt the gear under the hub

Step 5: Bolt the gear under the hub

Two M4 screws down through the bottom plate into each top block



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

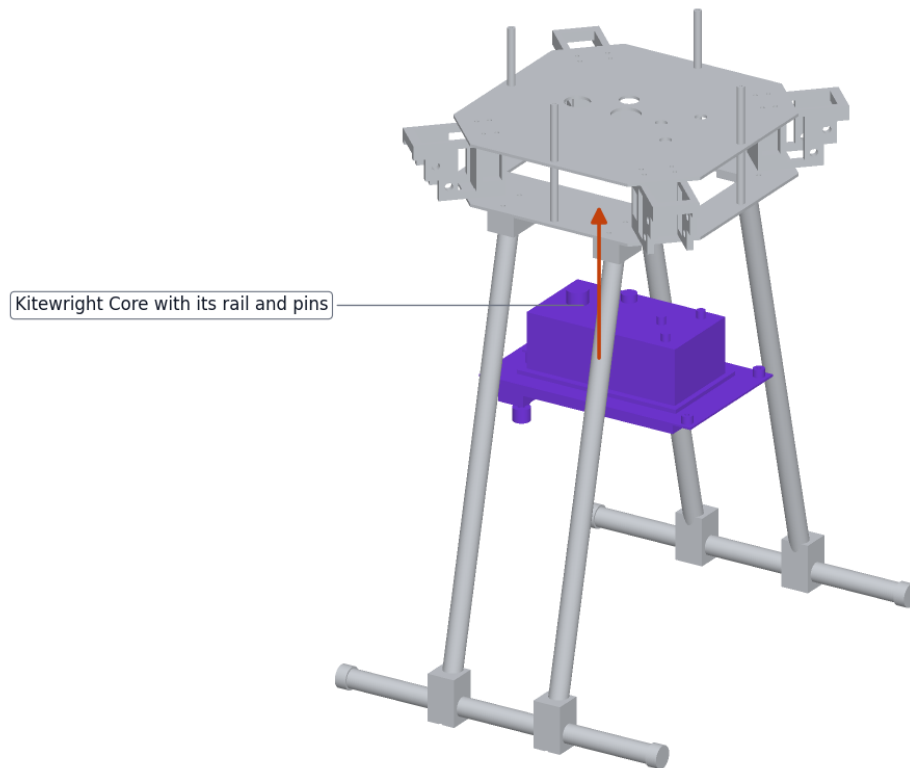
github.com/BoujeeEnjinia1701/kitewright-lift

Two M4 screws down through the bottom plate into each top block.

Step 6: Hang the Kitewright Core under the hub

Step 6: Hang the Kitewright Core under the hub

Lid up through the 200 x 112 opening; four M4 screws through the corner spacers; rail and pins come fitted



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

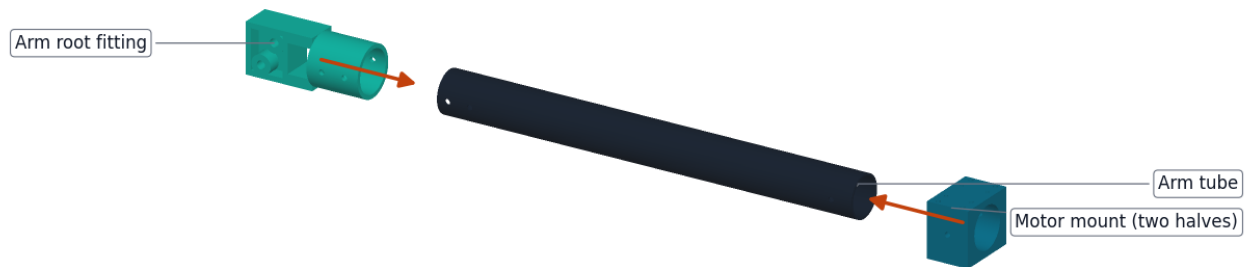
github.com/BoujeeEnjinia1701/kitewright-lift

From below, push the Core's lid up through the 200 x 112 mm opening until its four corner spacers meet the bottom plate; four M4 screws down through the plate into them. The rail, locking pins and pigtail come fitted. Check that the mast boss, SMA bulkheads and switch sit in their holes in the top plate without touching. Route the eight controller leads to the Core.

Step 7: Make up each arm

Step 7: Make up each arm

Bond and cross-bolt the tube into the root collar and the motor clamp; leave 24 h to cure



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

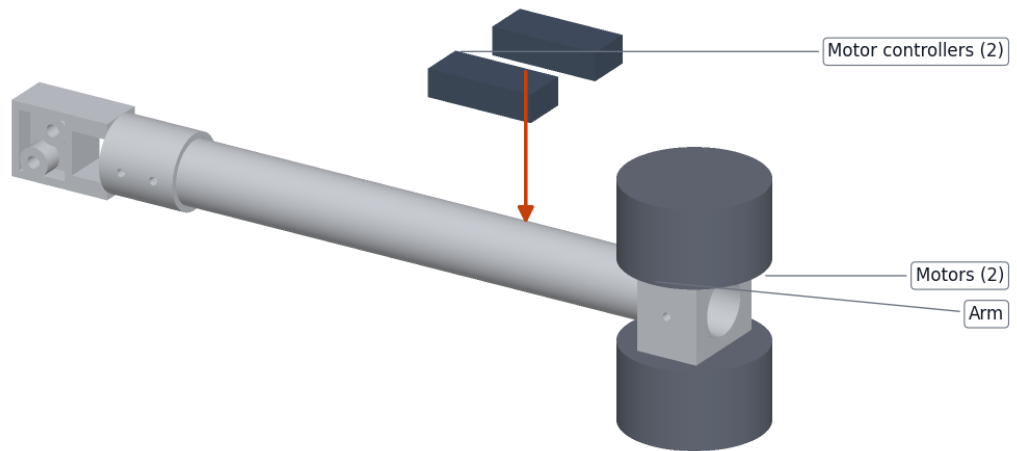
github.com/BoujeeEnjinia1701/kitewright-lift

Bond the tube 45 mm into the root fitting's collar and 40 mm into the motor clamp, with the clamp's top face square to the tongue's top. Fit the two M5 bolts through collar and tube and the M5 bolt through clamp and tube; close the clamp's four M4 bolts. Cure 24 h. Make four.

Step 8: Fit the motors and controllers

Step 8: Fit the motors and controllers

Upper motor on top, lower motor underneath, four M4 each; controllers strapped beside the tube



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

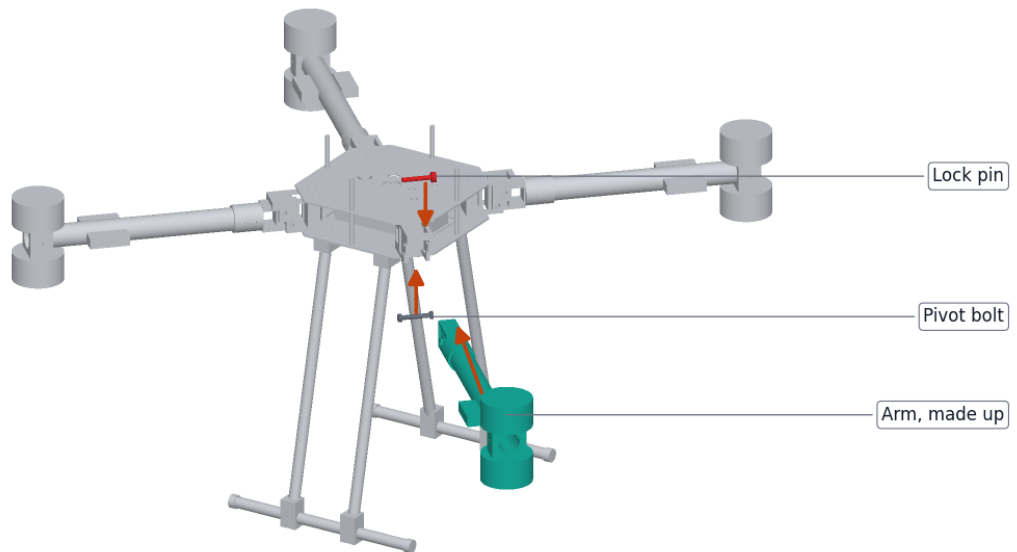
github.com/BoujeeEnjinia1701/kitewright-lift

Upper motor on the clamp's top face and lower motor, upside down, on its bottom face, four M4 screws each. Strap a controller to each side of the tube near the tip and solder or plug the motor leads. Leave a 60 mm slack loop in the two power leads where they will cross the hinge.

Step 9: Hang each arm in its hinge block

Step 9: Hang each arm in its hinge block

Tongue into the slot, pivot bolt through, nut snug; swing up, push the lock pin home



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

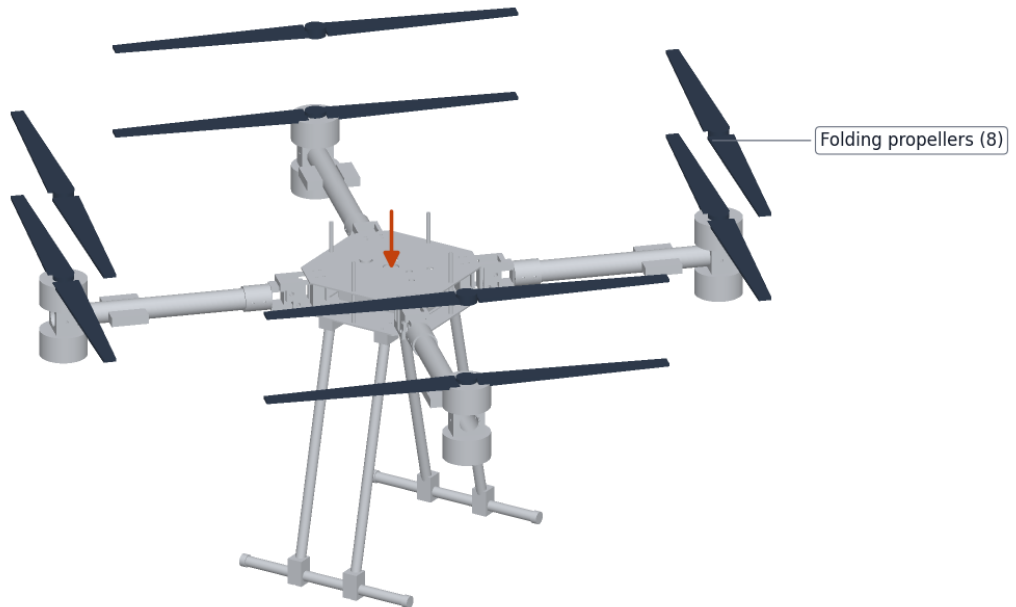
github.com/BoujeeEnjinia1701/kitewright-lift

Slide the tongue into the slot, push the pivot bolt through block and tongue, and snug the nyloc nut so the arm swings without play. Swing the arm up until its tongue meets the stop bridge and push the lock pin home. **Hold point:** each hinge is proof-loaded before first flight (section 6).

Step 10: Fit the propellers

Step 10: Fit the propellers

Upper and lower blades turn opposite ways; check the arrow on each hub



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

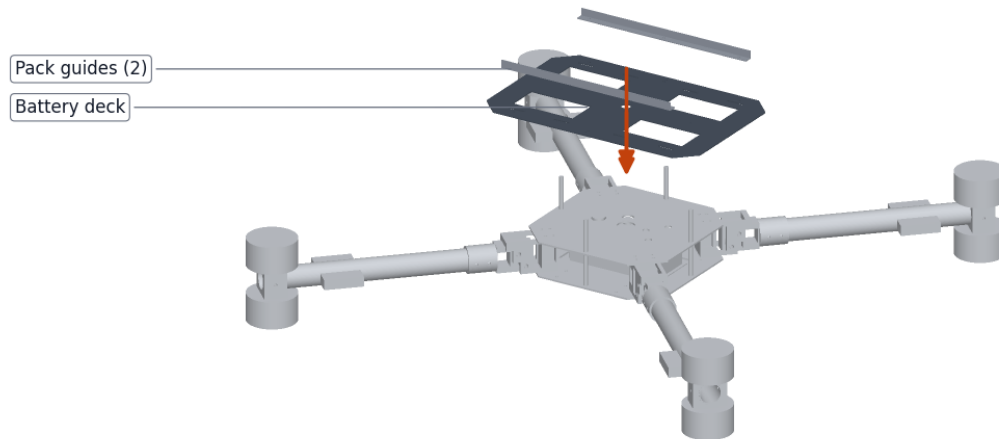
github.com/BoujeeEnjinia1701/kitewright-lift

Check the direction arrow on each hub against the motor's direction: upper and lower rotors on an arm turn opposite ways. **Hold point:** propellers stay off until the first power-up in section 5 is complete.

Step 11: Fit the battery deck and pack guides

Step 11: Fit the battery deck and pack guides

Deck on the four 60 mm standoffs with M3 screws; guides on M4 screws, upright legs outboard



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

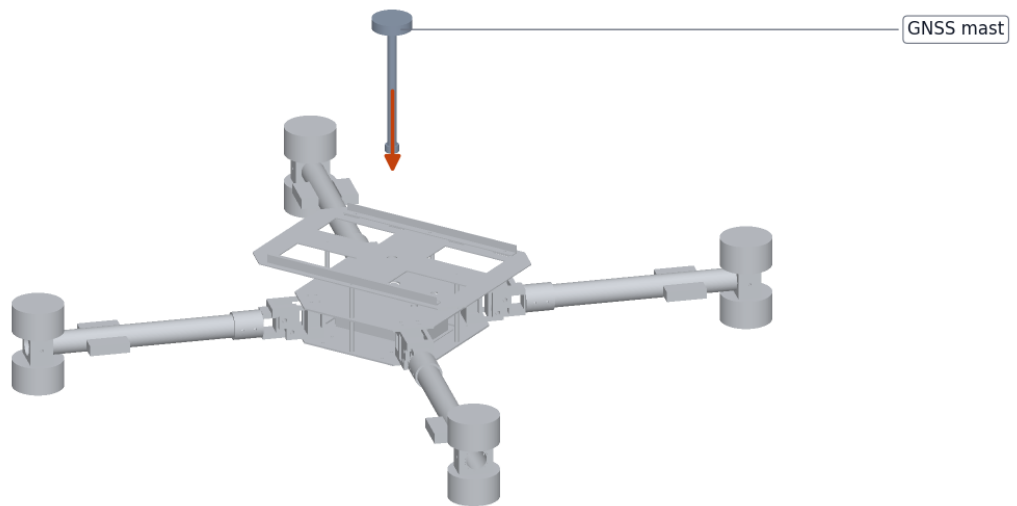
github.com/BoujeeEnjinia1701/kitewright-lift

Deck on the four 60 mm standoffs with M3 screws; guides on M4 screws, upright legs outboard; thread the four straps through their slots.

Step 12: Fit the GNSS mast and antenna leads

Step 12: Fit the GNSS mast and antenna leads

Mast foot on the deck centre with the Core's GNSS on top; the Core's antennas on extension leads to the gear struts



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

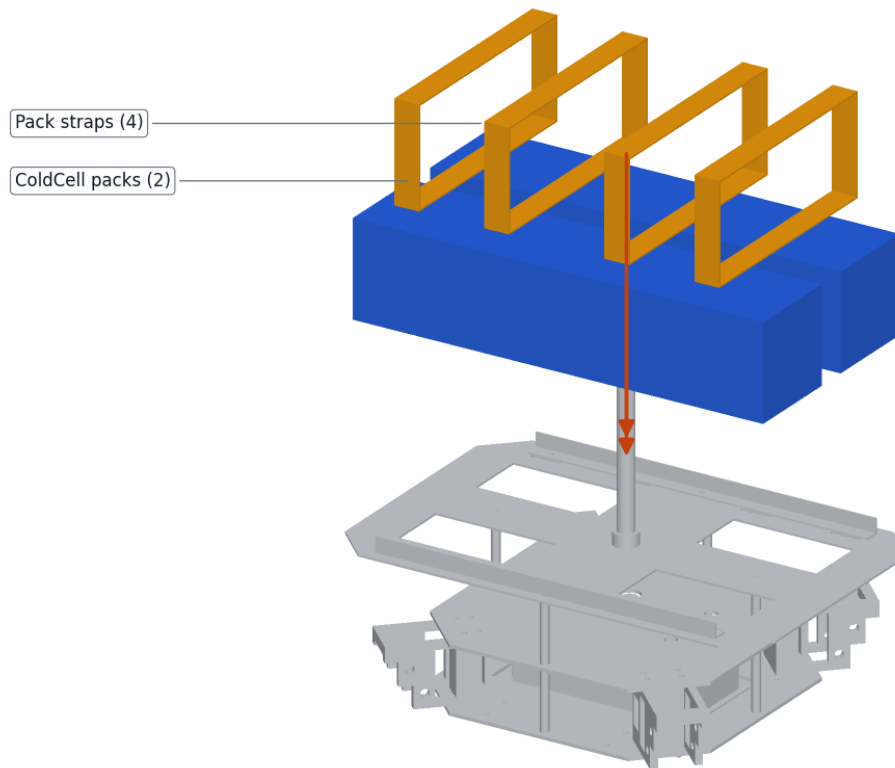
github.com/BoujeeEnjinia1701/kitewright-lift

Mast foot on the deck centre with the Core's GNSS receiver on top; lead its cable down through the deck and top plate to the Core. Screw the three SMA extension leads onto the Core's bulkheads through the top plate holes and clip the antennas to the rear gear struts, pointing down.

Step 13: Fit the ColdCell packs

Step 13: Fit the ColdCell packs

Both packs side by side, leads to the rear, between the guides; four cam straps round both; AS150 leads into the frame leads



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

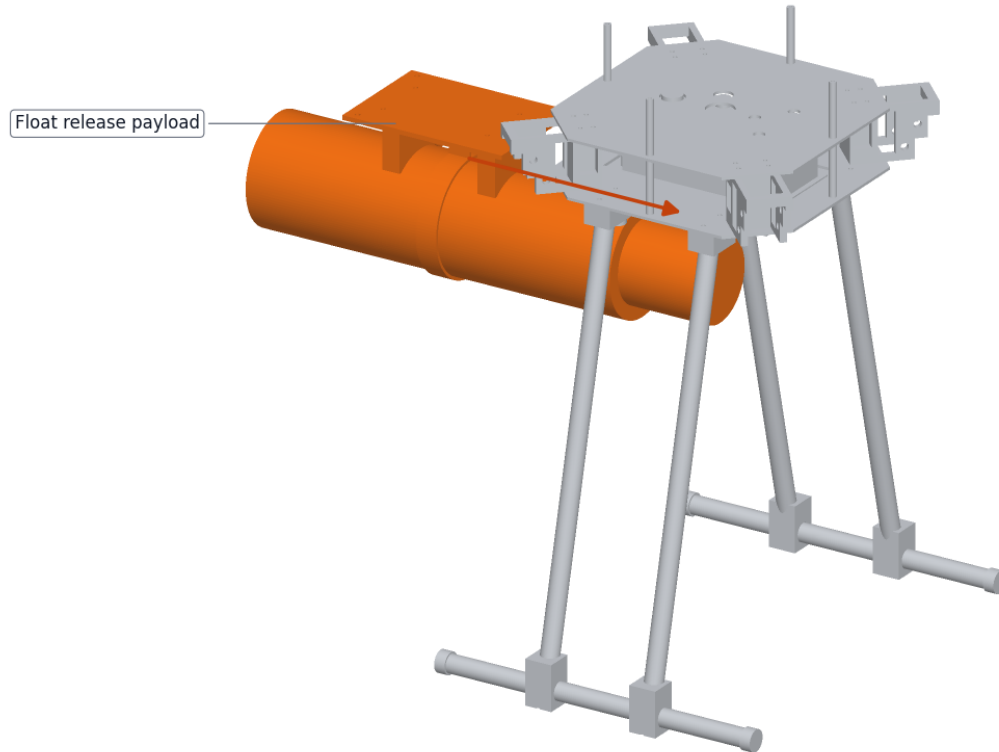
github.com/BoujeeEnjinia1701/kitewright-lift

Lay both packs side by side on the deck, leads to the rear, between the guides and either side of the mast; close the four straps round both; plug each into its AS150 lead from the Core. **Hold point:** packs go on only after the safety stop for first power (section 6).

Step 14: Slide in the float release payload

Step 14: Slide in the float release payload

Shoe in from the rear until it meets the front stops; both pins drop in; plug the Core's pigtail into the payload



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

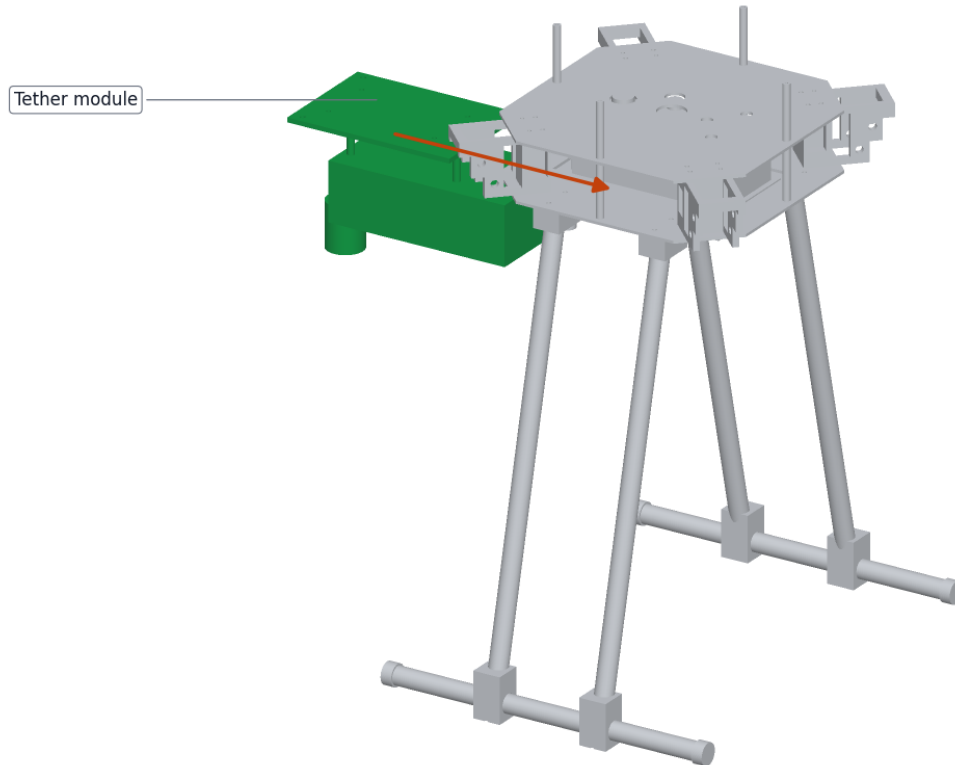
github.com/BoujeeEnjinia1701/kitewright-lift

From the rear, slide the payload shoe onto the rail lips until it meets the front stops; both locking pins drop into its holes (check both red bands are hidden). Plug the Core's DS-014 pigtail into the payload's socket in front of the shoe's notch.

Step 15: Or slide in the tether module

Step 15: Or slide in the tether module

Same shoe and pins; converter lead to the Core bus tether input; tether through the breakaway



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

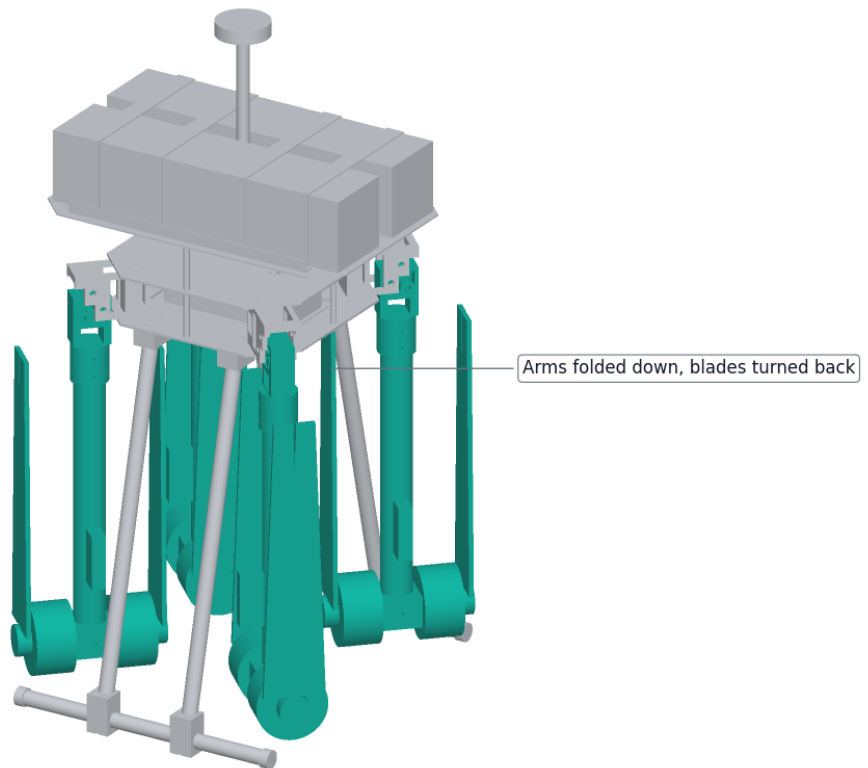
github.com/BoujeeEnjinia1701/kitewright-lift

The same rail and pins. Plug the converter's output lead into the Core bus tether input and lead the tether through the breakaway connector.

Step 16: Fold for transport

Step 16: Fold for transport

Payload off; pull each lock pin, swing the arm down; turn both blades back along the arm



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-lift

Take the payload off. For each arm, pull the lock pin, swing the arm down 90 degrees about its pivot, and turn both blades of each propeller back along the arm. Unfold in reverse; every lock pin goes back in and flagged before arming.

■ 5. First checks

Check	Requirement	How	Pass when
Weigh ready to fly, then with each payload	R1, R2	Hanging scale	25 kg or less with each payload; the rated payload confirmed (3.2 kg on paper)
Fold and pack	R9	Fold, place in a 1.2 x 0.6 x 0.5 m case, unfold and time the set-up	Fits the case; set-up in 10 min or less
Hinge proof load	R5 (safety)	Each arm on CalRig at 117 N m (1.5 times full thrust) for 1 min	No movement at the stop, no crack, no yield
First power-up, propellers off	R10	Core on the bench supply, then the packs; every motor turns the right way at low throttle	Each motor turns the right way; no fault on any controller
Motor and propeller thrust	R3	AltiRig at sea-level and 5,000 m density	Full-throttle thrust per motor about 10 kg at sea level
Tethered hover on the ground set	R7	Ground supply, insulation monitor, tether at 10 m height first	Bus steady at 51 V; no insulation alarm
Motor-cut test	R5	One motor cut by the autopilot at 10 m over cleared ground	Controlled descent and landing
Hover time	R4	Timed hover with the rated payload, logged energy and pack temperature	Time recorded against R4
Float drop	R8	Ten drops from 10 m onto a marked target on water	Float lands within the R8 distance
Wind hold	R6	Logged position hold on a windy day	Position held within the R6 distance

■ 6. Safety stops

- 1. Before any machining or cutting of carbon.** Dust extraction or a wet cut and a dust mask; carbon dust is an irritant and conducts electricity, so keep it away from the electronics.
- 2. Before the first power-up.** Propellers off. The Core is powered from a current-limited bench supply first. Packs are charged and handled to ColdCell's rules for lithium-ion packs: charging is blocked below 5 °C, packs are charged only in the fire-resistant charging box on a non-flammable surface, with a lithium-rated extinguisher or sand to hand. The frame-to-Core leads are checked for polarity and insulation before the first pack is plugged in.
- 3. Before the first flight.** Every hinge proof-loaded on CalRig (section 5) and inspected; every lock pin in and flagged; propellers balanced and checked for cracks; failsafes for loss of link and low battery set and tested; nobody within 15 m; a written arming and keep-out procedure followed.
- 4. Before energising the tether.** Ground supply earthed, insulation monitor and residual-current protection working, emergency stop tested, the tether laid clear of people, roads, water currents and power lines, and the breakaway fitted at the aircraft. Nobody touches the tether or the module while the supply is on. First tethered hovers at 10 m; never above 50 m.

5. **Before any payload drop.** Nobody under the aircraft except the person the float is for, who is approached from at least 10 m up; the float's line is tied to the float only. The release is tested on the ground first, with the aircraft unpowered.
6. **Before the motor-cut test.** Done over cleared ground at 10 m with a pilot ready to take over, and only after every other first check passes.
7. **After any hard landing.** Packs removed and watched outside for 24 h; hinges, bridges, pivots and arm tubes inspected before the next flight.

■ 7. Tools, skills and workspace

- **Tools:** drill press and a wooden V block; hand drill; fine abrasive cut-off disc or a tube cutter for carbon; files and deburring tool; M3, M4, M5 and M8 hex keys and nut drivers; torque screwdriver; 3D printer (or a printing service); sewing machine for webbing (or a sail or upholstery shop); soldering station for the motor leads; hanging scale to 50 kg; multimeter; insulation tester for the tether set.
- **Bought machining:** the hinge blocks, root fittings, motor clamps and gear blocks are best made by a machine shop or an online CNC service from the making sketches; the carbon plates by a CNC routing service.
- **Skills:** careful drilling and bonding of carbon; multirotor assembly and autopilot setup (PX4 or ArduPilot); handling lithium packs; for the tether set, an electrician who works with DC above 120 V.
- **Workspace:** a clean bench with dust extraction for carbon work; a non-flammable area for pack charging; a clear outdoor flight area with permission, at least 30 m across, for the first flights.

■ 8. Where the numbers come from

- `cad/src/model.py` : the parametric model and its constructability checks; STEP and STL in `cad/step` and `cad/stl`.
- `cad/drawings/KWL-DWG-001` (general arrangement), `KWL-DWG-002` (folded) and `KWL-DWG-101` to `KWL-DWG-114` (making sketches).
- `docs/04-calcs/01-sizing.md` and `docs/04-calcs/sizing.py` (KWL-CAL-001).
- `bom/bom.csv` (parts, specifications and indicative prices).
- `docs/decisions/0001-trl2-review-decisions.md` and `docs/decisions/0002-design-for-construction.md`.
- `cad/src/build_plan_media.py` (every picture in this plan).